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Liver Transplantation

Number: 0596

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Scope of Policy

This Clinical Policy Bulletin addresses liver transplantation.

I. **Medical Necessity**

A. Aetna considers liver transplantation medically necessary for the indications listed below in Section I.B. for *the following*:

1. Adolescents 12 years of age or older and adults with *either*:

   a. A Model of End-stage Liver Disease (MELD) score (see Appendix) greater than 10; *or*
   b. Who are approved for transplant by the United Network for Organ Sharing (UNOS) Regional Review Board; *or*
   c. Who meet the transplant institution's selection criteria; *and*

2. Children less than 12 years of age who meet the transplanting institution's selection criteria.

In the absence of an institution's selection criteria, requests for liver transplantation are subject to medical necessity review for children, and for adolescents and adults with a MELD score of 10 or less who have not been approved by the UNOS Regional Review Board.

B. **Medically Necessary Indications (not an all-inclusive list)**

Aetna considers orthotopic (normal anatomical position) liver transplantation (with cadaveric organ, reduced-size organ, living related organ, and split liver) medically necessary for members with end-stage liver disease (ESLD) due to *any* of the following conditions who meet medical necessity criteria in Section I.A.:

1. **Cholestatic diseases**

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a. Biliary atresia;
b. Familial cholestatic syndromes;
c. Primary biliary cirrhosis;
d. Primary sclerosing cholangitis with development of secondary biliary cirrhosis;

2. **Hepatocellular diseases**

a. Alcoholic cirrhosis;
b. Chronic active hepatitis with cirrhosis (hepatitis B or C);
c. Cryptogenic cirrhosis;
d. Idiopathic autoimmune hepatitis;
e. Post-necrotic cirrhosis due to hepatitis B surface antigen negative state;

3. **Malignancies**

a. Primary hepatocellular carcinoma confined to the liver when *all* of the following criteria are met:

i. Any lung metastases that have been shown to be responsive to chemotherapy; *and*
ii. Member is not a candidate for subtotal liver resection; *and*
iii. Member meets UNOS criteria for tumor size and number; *and*
iv. There is no identifiable extra-hepatic spread of tumor to surrounding lymph nodes, abdominal organs, bone or other sites; *and*
v. There is no macrovascular involvement;

Note: These criteria are intended to be consistent with UNOS guidelines for selection of liver transplant candidates for hepato-cellular carcinoma (HCC).

b. Hepatoblastomas in members less than 12 years of age when *all* of the following criteria are met:

i. Member is not a candidate for subtotal liver resection; *and*
ii. Member meets UNOS criteria for tumor size and number; *and*
iii. There is no identifiable extra-hepatic spread of tumor to surrounding lungs, abdominal organs, bone or other sites; **Note**: Spread of hepatoblastoma to veins and lymph nodes does not disqualify a member for coverage of a liver transplant;

c. Epithelioid hemangioendotheliomas;
d. Intra-hepatic cholangiocarcinomas (i.e., cholangiocarcinomas confined to the liver);
e. Large, unresectable fibrolamellar HCCs;
f. Metastatic neuroendocrine tumors (carcinoid tumors, apudomas, gastrinomas, glucagonomas) in persons with severe symptoms and with metastases restricted to the liver, who are unresponsive to adjuvant therapy after aggressive surgical resection including excision of the primary lesion and reduction of hepatic metastases;

4. **Vascular diseases**

a. Budd-Chiari syndrome;
b. Veno-occlusive disease;

5. **Metabolic disorders and metabolic liver diseases with cirrhosis (not an all-inclusive list)**

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a. Alpha 1-antitrypsin deficiency;
b. Hemochromatosis;
c. Inborn errors of metabolism;
d. Protoporphyria;
e. Wilson's disease;

6. **Miscellaneous**

a. Familial amyloid polyneuropathy;
b. Polycystic disease of the liver;
c. Porto-pulmonary hypertension (pulmonary hypertension associated with liver disease or portal hypertension) in persons with a mean pulmonary artery pressure by catheterization of less than 35 mm Hg;
d. Toxic reactions (fulminant hepatic failure due to mushroom poisoning, acetaminophen (Tylenol) overdose, etc.);
e. Trauma;
f. Hepato-pulmonary syndrome when the following selection criteria are met:

i. Arterial hypoxemia (PaO_2 less than 60 mm Hg or AaO_2 gradient greater than 20 mm Hg in supine or standing position); *and*
ii. Chronic liver disease with non-cirrhotic portal hypertension; *and*
iii. Intrapulmonary vascular dilatation (as indicated by contrast-enhanced echocardiography, technetium-99 macroaggregated albumin perfusion scan, or pulmonary angiography).

C. **Retransplantation**

Aetna considers retransplantation following a failed liver transplant medically necessary if the initial transplant was performed for a covered indication.

D. **Compatibility Testing**

Aetna considers compatibility testing of prospective liver donors medically necessary for living donor liver transplantation. **Note:** Compatibility testing and related services of the prospective live donor, as well as inpatient care of a compatible live donor for the donation procedure, may be covered under the member's medical benefit.

E. **Contraindications**

Aetna considers liver transplantation not medically necessary for members with *any* of the following absolute contraindications to liver transplantation:

1. Active sepsis outside the biliary tract;
2. Inability to adhere to the regimen necessary to preserve the transplant including homelessness, active substance abuseFootnote1* (e.g., alcohol, cocaine, crystal meth, heroin, methadone, and/or narcotics, etc.), and/or unstable psychiatric disease/ psychosocial problem"
3. Other effective medical treatments or surgical options are available;
4. Presence of significant organ system failure other than kidney, liver or small bowel.

Footnote1* Exception to this contraindication will be allowed with supporting documentation (within 4 weeks) demonstrating 3 months of stability from treating addiction medical professional or psychiatrist.

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I. Experimental, Investigational, or Unproven

The following interventions are considered experimental, investigational, or unproven because their safety and effectiveness has not been established:

- Basiliximab for induction immunosuppression in individuals undergoing liver transplantation (LT);
- Bioartificial liver transplantation;
- Biomarkers (acid labile nitroso-compounds (NOx), serum amyloid A protein, procalcitonin, peripheral blood T-cell activation, interleukin 2 (IL-2) receptor, guanylate-binding protein-2 mRNA, graft-derived cell-free DNA, pi-glutathione S-transferase, alpha-glutathione S-transferase and serum HLA class I soluble antigens) for diagnosis of acute allograft rejection following liver transplantation;
- Ectopic or auxiliary liver transplantation;
- Everolimus to prevent organ rejection after liver transplantation;
- Factor V Leiden and F2 testing for member scheduled to receive partial liver transplant for primary sclerosing cholangitis;
- Hepatocellular (hepatocyte) transplantation;
- Hypothermic machine perfusion for reduction of the incidences of early allograft dysfunction and biliary complications after LT;
- Liver elastography and Doppler examination of the portal veins and hepatic arteries for routine yearly surveillance following liver transplantation;
- Liver transplantation for malignancies other than those listed as covered above;
- Liver transplantation for the treatment of extra-hepatic hilar cholangiocarcinoma;
- Measurements of plasma and urinary neutrophil gelatinase-associated lipocalin (NGAL) for predicting acute kidney injury following orthotopic liver transplantation;
- Molecular Adsorbent Recirculating System (MARS) for the treatment of progressive familial intrahepatic cholestasis;
- Normothermic machine perfusion of donor liver;
- OrganOx metra System for transportation and preservation of the liver prior to transplantation;
- Peri-operative use of vasopressin in liver transplantation;
- Peri-operative use of sorafenib in liver transplantation;
- Scaffold-based transplantation (combination of xeno-organ and cell transplantations) as an alternative for orthotopic LT;
- Testing of high mobility group box protein 1 (HMGB1) gene polymorphisms for prediction of morbidity and mortality after liver transplantation;
- Transient elastography for diagnosis of acute cellular rejection following liver transplantation;
- Ursodeoxycholic acid (UDCA), adjuvant use to prevent acute cellular rejection after liver transplantation;
- Xenotransplantation.

Note: For policy on hepatitis B immune globulin for prophylaxis of recurrent hepatitis B infection in HbsAg positive liver transplant recipients, see CPB 0544 - Immune Globulins for Post-Exposure Prophylaxis.

II. Related Policies

- CPB 0544 - Immune Globulins for Post-Exposure Prophylaxis

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Table:

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description

CPT codes covered if selection criteria are met:

47133	Donor hepatectomy (including cold preservation), from cadaver donor
47135	Liver allotransplantation; orthotopic; partial or whole, from cadaver or living donor, any age
47140	Donor hepatectomy (including cold preservation), from living donor; left lateral segment only (segments II and III)
47141	total left lobectomy (segments II, III and IV)
47142	total right lobectomy (segments V, VI, VII and VIII)
47143	Backbench standard preparation of cadaver donor whole liver graft prior to allotransplantation, including cholecystectomy, if necessary, and dissection and removal of surrounding soft tissues to prepare the vena cava, portal vein, hepatic artery, and common bile duct for implantation; without trisegment or lobe split
47144	with trisegment split of whole liver graft into two partial liver grafts (ie, left lateral segment (segments II and III) and right trisegment (segments I and IV through VIII))
47145	with lobe split of whole liver graft into two partial liver grafts (ie, left lobe (segments II, III, and IV) and right lobe (segments I and V through VIII))
47146	Backbench reconstruction of cadaver or living donor liver graft prior to allotransplantation; venous anastomosis, each
47147	arterial anastomosis, each

CPT codes not covered for indications listed in the CPB (not an all-inclusive list):

Molecular Adsorbent Recirculating System (MARS), Normothermic machine perfusion of donor liver, measurements of plasma and urinary neutrophil gelatinase-associated lipocalin (NGAL), acid labile nitroso-compounds (NOx), peripheral blood T-cell activation, guanylate-binding protein-2 mRNA, graft-derived cell-free DNA and serum HLA class I soluble antigens, Pi-glutathione S-transferase and Alpha-glutathione S-transferase, Serum amyloid A protein, scaffold-based transplantation, OrganOx metra System, testing of high mobility group box protein 1 (HMGB1) gene polymorphisms **- no specific code**

0894T	Cannulation of the liver allograft in preparation for connection to the normothermic perfusion device and decannulation of the liver allograft following normothermic perfusion
0895T	Connection of liver allograft to normothermic machine perfusion device, hemostasis control; initial 4 hours of monitoring time, including hourly physiological and laboratory assessments (eg, perfusate temperature, perfusate pH, hemodynamic parameters, bile production, bile pH, bile glucose, biliary bicarbonate, lactate levels, macroscopic assessment)
0896T	Connection of liver allograft to normothermic machine perfusion device, hemostasis control; each additional hour, including physiological and laboratory assessments (eg, perfusate temperature, perfusate pH, hemodynamic parameters, bile production, bile pH, bile glucose, biliary bicarbonate, lactate levels, macroscopic assessment) (List separately in addition to code for primary procedure)
76391	Magnetic resonance (eg, vibration) elastography
76981	Ultrasound, elastography; parenchyma (eg, organ)

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Code	Code Description
81240	F2 (prothrombin, coagulation factor II) (eg, hereditary hypercoagulability) gene analysis, 20210G>A variant
81241	F5 (coagulation Factor V) (eg, hereditary hypercoagulability) gene analysis, Leiden variant
81405	Molecular pathology procedure, Level 6 [interleukin 2 (IL-2) receptor]
84145	Procalcitonin
91200	Liver elastography, mechanically induced shear wave (eg, vibration), without imaging, with interpretation and report
93975	Duplex scan of arterial inflow and venous outflow of abdominal, pelvic, scrotal contents and/or retroperitoneal organs; complete study

Other CPT codes related to the CPB:

47120 - 47130	Hepatectomy, resection of liver; partial lobectomy; trisegmentectomy; total left lobectomy; or total right lobectomy
81370 – 81383	HLA Class I and II typing, low resolution (eg, antigen equivalents)
86826 – 86828	Human leukocyte antigen (HLA) crossmatch, non-cytotoxic (eg, using flow cytometry)
86902	Blood typing, serologic; antigen testing of donor blood using reagent serum, each antigen test

HCPCS codes not covered for indications listed in the CPB:

B4155	Enteral formula, nutritionally incomplete/modular nutrients, includes specific nutrients, carbohydrates (e.g., glucose polymers), proteins/amino acids (e.g., glutamine, arginine), fat (e.g., medium chain triglycerides) or combination, administered through an enteral feeding tube, 100 calories = 1 unit
J0480	Injection, basiliximab, 20 mg
J2596	Injection, vasopressin (long grove), not therapeutically equivalent to j2598, 1 unit
J2598	Injection, vasopressin, 1 unit
J2599	Injection, vasopressin (american regent) not therapeutically equivalent to j2598, 1 unit
J2601	Injection, vasopressin (baxter), 1 unit

ICD-10 codes covered if selection criteria are met:

B16.0, B16.2, B18.0 - B18.1, B19.11	Acute hepatitis B with hepatic coma
B16.1, B16.9, B19.10	Acute hepatitis B without mention of hepatic coma
B17.10	Acute hepatitis C without hepatic coma
B17.11	Acute hepatitis C with hepatic coma
B18.2	Chronic viral hepatitis C
B19.20 - B19.21	Unspecified viral hepatitis C
C22.0	Liver cell carcinoma
C22.1	Intrahepatic bile duct carcinoma

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Code	Code Description
C22.2	Hepatoblastoma [in children]
D37.6	Neoplasm of uncertain behavior of liver, gallbladder and bile ducts [Epithelioid hemangioendotheliomas]
E70.0 - E72.9	Disorders of aromatic amino-acid metabolism, disorders of branched-chain amino-acid metabolism and fatty-acid metabolism and other disorders of amino-acid metabolism
E80.0	Hereditary erythropoietic porphyria [Erythropoietic protoporphyria]
E83.01	Wilson's disease
E83.10, E83.19	Other and unspecified disorders of iron metabolism
E83.110 - E83.119	Hereditary hemochromatosis
E85.1	Neuropathic heredofamilial amyloidosis
E88.01	Alpha-1-antitrypsin deficiency
I82.0	Budd-Chiari syndrome
K70.2	Alcoholic fibrosis and sclerosis of liver
K70.30 - K70.31	Alcoholic cirrhosis of liver
K73.1 - K73.8	Chronic hepatitis, not elsewhere classified
K74.3	Primary biliary cirrhosis
K74.4	Secondary biliary cirrhosis
K74.69	Other cirrhosis of liver [cryptogenic cirrhosis (of liver)] [post-necrotic cirrhosis (of liver)]
K75.4	Autoimmune hepatitis
K75.81 - K75.9	Other and unspecified inflammatory liver diseases
K76.5	Hepatic veno-occlusive disease
K76.81	Hepatopulmonary syndrome
K83.01 - K83.09	Cholangitis [primary sclerosing cholangitis with development of secondary biliary cirrhosis]
K83.1	Obstruction of bile duct [MARS]
Q44.2	Atresia of bile ducts
Q44.3	Congenital stenosis and stricture of bile ducts
Q44.6	Cystic disease of liver
T86.40 - T86.49	Complications of liver transplant
Z52.6	Liver donor

ICD-10 codes not covered for indications listed in the CPB:

A41.9	Sepsis, unspecified organism
C24.0	Malignant neoplasm of extrahepatic bile duct
T86.41	Liver transplant rejection
Z76.82	Awaiting organ transplant status [liver]
Z94.4	Liver transplantation status [not covered for the use of everolimus to prevent organ rejection]

ICD-10 codes contraindicated for this CPB:

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Code	Code Description
F01 – F99	Mental, Behavioral and Neurodevelopmental disorders [active substance abuse, unstable psychiatric disease, psychosocial problem]
Z59.00-Z59.02	Homelessness
Z59.811	Housing instability, housed, with risk of homelessness
Z59.812	Housing instability, housed, homelessness in past 12 months

Background

Progressive liver diseases that result in death either in short-term or long-term is known as end-stage liver disease (ESLD), which is evidenced by irreversible, progressive liver dysfunction, variceal bleeding, encephalopathy, synthetic dysfunction, poor growth, or poor nutritional status. The most common causes of ESLD include infection (e.g., acute or chronic hepatitis), toxic effects (e.g., alcohol, medications), disorders of metabolism (e.g., hemochromatosis, Wilson's disease), tumors (primary or metastatic), and malformations (e.g., primary biliary atresia). Liver transplantation is an effective treatment for fulminant (acute) hepatic failure and for many chronic liver diseases.

A liver transplant is usually positioned in the normal anatomical position (orthotopic) following a total hepatectomy of the recipient. In auxiliary liver transplantation, a second liver is implanted ectopically and the recipient's own liver remains in-situ. A major concern of ectopic transplantation is the recipient's diseased liver may harbor bacterial, fungal or viral infection or cancer. Advances in surgical techniques and immunosuppressive drugs have resulted in increased survival rates (with 1-year survival rates in the 85 to 90 % range, and 5-year survival rates exceeding 70 %). Currently, 10 to 20 % of liver transplanted patients are retransplanted with a success rate of greater than 50 %.

Hepatitis C cirrhosis is the most common indication for liver transplantation. Alcoholic liver disease remains a controversial indication for liver transplantation but carefully selected patients do well. Some of the common indications for liver transplantation are as follows:

1. Alcoholic liver disease (after a period of abstinence)
2. Chronic active hepatitis (usually secondary to hepatitis B and C)
3. Cryptogenic cirrhosis
4. Primary biliary cirrhosis
5. Primary sclerosing cholangitis.

Hepato-cellular carcinoma (HCC) complicates many chronic liver diseases. However, a small tumor is not a contraindication to transplantation since tumor rarely recurs in these patients. In contrast, most patients with large (greater than 5 cm in diameter) or multiple hepatomas or most other types of cancer are not considered for transplantation since tumors recur rapidly. At present, there is insufficient evidence that liver transplantation is an effective treatment for other malignancies that affect the liver such as metastatic disease, bile duct carcinoma, and epitheloid hemangioendothelioma, among others. An assessment by the Agency for Healthcare Research and Quality (Beavers et al, 2001) on liver transplantation for malignancies other than HCC concluded that “[t]he available evidence does not provide a clear profile of patients who might be optimal candidates for such therapy.” Contraindications to liver transplantation include extra-hepatic malignancy, severe cardiopulmonary disease, systemic sepsis, and an inability to comply with regular pharmacotherapy.

Liver transplantation is an effective treatment for a variety of acute and chronic diseases of the liver in the pediatric (less than 18 years of age) population. Approximately 15 % of the liver transplantations performed yearly in the United States are in pediatric patients. Most children who need liver transplantation are young (age less than 3 years) and small (body weight less than 45 pounds). Size-matched organs are given

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preference in organ allocation. However, because of the severe scarcity of pediatric donor livers, techniques such as reduced size ("cut down") and split (a liver is split between 2 recipients) liver transplantations are used to reduce the size of adult donor livers to fit pediatric recipients. Donation of the left lobe of the liver by a living adult relative ("living related donor") is also an option. Liver transplantation in children is indicated for ESLD from any etiology in the absence of contraindications. The most common indication for pediatric liver transplantation is biliary atresia, often after failure to respond to a porto-enterostomy. In addition, unresectable tumors and liver-based metabolic deficiencies may be indications for liver transplantation.

The Model for End-Stage Liver Disease (MELD) is a numerical scale, ranging from 6 (less ill) to 40 (gravely ill), that is used for adult liver transplant candidates. It gives each individual a 'score' (number) based on how urgently he or she needs a liver transplant within the next 3 months. The number is calculated by a formula using bilirubin, prothrombin time, and creatinine. Candidates under the age of 12 are placed in categories according to the Pediatric End-stage Liver Disease (PELD) scoring system. PELD is similar to MELD but uses some different criteria to recognize the specific growth and development needs of children. PELD scores may also range higher or lower than the range of MELD scores. The PELD scoring system takes into account the patient's bilirubin, prothrombin time, albumin, growth failure, and whether the child is less than 1 year old. A liver transplantation is rarely necessary for persons with a MELD score of less than 10. According to data from the United Network for Organ Sharing (UNOS), of almost 5,000 liver transplants that were performed in 2002, only 181 transplants were performed on patients with a MELD score of less than 10.

The MELD/PELD score is a well-validated measure of short-term mortality from liver disease; however, referring physicians who believe a patient faces a greater mortality risk than predicted by the MELD/PELD score can request accelerated listing. UNOS Regional Review Boards can approve or deny these requests, and a study by Voight et al (2004) concluded that these boards fairly and accurately distinguish between high- and low-risk patients. The study found that the denials of physicians' requests for accelerated listings did not increase mortality for those patients. To determine the effect of UNOS Regional Review Board decisions on the mortality of physician-referred patients, investigators analyzed 1,965 nationwide referrals to UNOS Regional Review Boards. They noted which cases were approved and which were denied, and gathered information about patient deaths while awaiting transplantation. The investigators found that there was no significant difference in survival to transplantation whether accelerated listing was approved or denied for adult or pediatric cases. In addition, the researchers examined whether or not referring physicians predicted death better than the MELD/PELD score. The investigators found that the physicians had poor predictive capacity and added no additional information to to the risk assessment by the MELD/PELD score. The investigators concluded that the MELD-PELD score is a better predictor of mortality than the judgement of the referring physician, but the UNOS Regional Review Board process adds additional information (e.g., Voight et al, 2004).

Dimmock et al (2008) noted that deoxyguanosine kinase (DGUOK) deficiency is the commonest type of mitochondrial DNA depletion associated with a hepato-cerebral phenotype. These researchers assessed predictors of survival and therapeutic options in patients with DGUOK deficiency. A systematic search of MEDLINE, LILAC, and SCIELO was performed to identify peer-reviewed clinical trials, randomized controlled trials, meta-analyses, and other studies with clinical pertinence. Deoxyguanosine kinase deficiency was searched with the terms dGK, DGUOK, mitochondrial DNA depletion, mtDNA, and hepatocerebral. Bibliographies of identified articles were reviewed for additional references. A total of 13 identified studies met the inclusion criteria and were used in this study. The analysis revealed that DGUOK deficiency is associated with a variable clinical phenotype. Long-term survival is best predicted by the absence of profound hypotonia, significant psychomotor retardation, or nystagmus. In the presence of these features, there is increased mortality, and liver transplantation does not confer increased survival. The authors concluded that liver transplantation appears to be futile in the presence of specific neurological signs or symptoms in patients affected with DGUOK deficiency. Conversely, in the absence of these neurological features, liver transplantation may be considered a potential treatment.

Bioartificial Liver Transplantation

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Artificial and bioartificial livers have been developed for use as a bridge to transplant in patients with liver failure or to allow recovery in persons with acute liver failure. Liu et al (2004) reported on the results of a meta-analysis of 12 trials of artificial or bioartificial support systems versus standard medical therapy, involving 483 patients, and 2 trials comparing different artificial support systems, involving 105 patients. Most trials had unclear methodological quality. Compared to standard medical therapy, support systems had no significant effect on mortality (relative risk [RR] 0.86; 95 % confidence interval [CI]: 0.65 to 1.12) or bridging to liver transplantation (RR 0.87; 95 % CI: 0.73 to 1.05), but a significant beneficial effect on hepatic encephalopathy (RR 0.67; 95 % CI: 0.52 to 0.86). Subgroup analysis indicated that artificial and bioartificial livers may reduce mortality by 1/3 in acute-on-chronic liver failure (RR 0.67; 95 % CI: 0.51 to 0.90), but not in acute liver failure (RR 0.95; 95 % CI: 0.71 to 1.29). The authors noted that the incidence of adverse events was inconsistently reported. They concluded that, although artificial support systems may reduce mortality in acute-on-chronic liver failure, "considering the strength of the evidence additional randomised clinical trials are needed before any support system can be recommended for routine use."

More recently, Demetriou et al (2004) reported on the first prospective, randomized controlled trial of bioartificial liver, the HepatAssist Liver Support System in 171 patients with severe acute liver failure, including both fulminant/sub-fulminant hepatic failure and primary non-function following liver transplantation. For the entire patient population, survival at 30 days was 71 % for patients assigned to the bioartificial liver versus 62 % for patients in the control group ($p = 0.26$). After exclusion of primary non-function patients, survival was 73 % for persons assigned to the bioartificial liver versus 59 % for persons in the control group ($p = 0.12$). When survival was analyzed accounting for confounding factors, in the entire patient population, there was no difference between the 2 groups (risk ratio = 0.67; $p = 0.13$). However, differences in survival between bioartificial liver and control patients with fulminant/sub-fulminant hepatic failure reached marginal statistical significance (risk ratio = 0.56; $p = 0.048$). The authors concluded that this study demonstrated improved survival in patients with fulminant/subfulminant hepatic failure. These results would need to be confirmed in additional prospective randomized studies before conclusions can be drawn about the effectiveness of the bioartificial liver.

Peri-Operative Use of Sorafenib

Qi and colleagues (2015) examined if the application of sorafenib during the peri-operative period of liver transplantation (LT) improves prognosis in liver cancer patients. These investigators searched PubMed, EMBASE and MEDLINE for eligible articles. A total of 4 studies were found that fulfilled the previously agreed-upon standards. They then performed a systematic review and meta-analysis on the enrolled trials that met the inclusion criteria. Out of the 104 studies identified in the database, 82 were not clinical experiments, and 18 did not fit the inclusion standards. Among the remaining 4 articles, only 1 was related to the pre-operative use of sorafenib, whereas the other 3 were related to its post-operative use. As the heterogeneity among the 4 studies was high, with an I(2) of 86 %, a randomized effect model was applied to pool the data. The application of sorafenib before LT had a hazard ratio (HR) of 3.29 (95 % CI: 0.33 to 32.56). The use of sorafenib after LT had an HR of 1.44 (95 %CI: 0.27 to 7.71). The overall pooled HR was 1.68 (95 %CI: 0.41 to 6.91). The authors concluded that the results showed that the use of sorafenib during the peri-operative period of LT did not improve patient survival significantly. In fact, sorafenib could even lead to a worse prognosis, as its use may increase the hazard of poor survival.

Mancuso et al (2015) stated that data on survival and safety of sorafenib for hepatocellular carcinoma recurrence after LT are still equivocal. These researchers performed a meta-analysis of published studies, with the aim of estimating the 1-year rates of survival, analyzing the variability in survival rates and, finally, identifying the factors associated with a longer survival. Data from 8 of the 17 selected studies were pooled, while the other 9 were excluded because survival rates were missing. All included studies were retrospective. Overall, the 1-year survival ranged from 18 % to 90 %. Tumor progression was the main cause of death. The second cause was bleeding, reported only in patients undergoing m-Tor inhibitor therapy. The pooled estimate of 1-year survival was 63 %. There was a significant heterogeneity among studies ($p < 0.0001$). Among the 34 variables assessed by univariate meta-regression, 5 were associated with an increase in the 1-year survival rate:

I. male gender ($p = 0.001$);

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II. time to progression (p = 0.038); and adverse drug events, divided in
III. gastrointestinal (p = 0.038),
IV. cardiovascular (p = 0.029), and
V. dermatological (p = 0.014).

I. male gender (p = 0.001);
II. time to progression (p = 0.038); and adverse drug events, divided in
III. gastrointestinal (p = 0.038),
IV. cardiovascular (p = 0.029), and
V. dermatological (p = 0.014).

The authors concluded that additional data from multi-center prospective studies are needed to clearly determine if sorafenib is a safe and acceptable treatment in hepatocellular carcinoma recurrence after LT. Nevertheless, its association with m-Tor inhibitors should be discouraged.

An UpToDate review on "Long-term management of adult liver transplant recipients" (Gaglio and Cotler, 2015) does not mention the use of sorafenib as a management agent.

Split Liver Transplantation versus Whole Liver Transplantation

Wan and colleagues (2015) noted that split liver transplantation (SLT) has proven to be an effective technique to reduce the mortality of children on the waiting list, but whether creating 2 split grafts from 1 standard-criteria whole liver would compromise outcomes of adult recipients remains uncertain. These investigators conducted this meta-analysis to compare outcomes of right lobe SLT and whole liver transplantation (WLT) in adult patients. PubMed, Embase, and the Cochrane Library were searched for relevant articles published before December 2014. Outcomes assessed were patient survival, graft survival and major surgical complications after transplantation. Pooled odds ratios (OR) with 95 % CI were calculated to synthesize the results. A total of 17 studies with a total of 4,8457 patients met the full inclusion criteria. Patient survival and graft survival rates were all found to be equivalent between SLT and WLT recipients. However, SLT was associated with higher rates of overall biliary complications (OR = 1.66, 95 % CI: 1.29 to 2.15, p < 0.001), bile leaks (OR = 4.30, 95 % CI: 2.97 to 6.23, p < 0.001), overall vascular complications (OR = 1.81, 95 % CI:1.29 to 2.53, p < 0.001), hepatic artery thromboses (OR = 1.71, 95 % CI: 1.17 to 2.50, p = 0.005) and outflow tract obstructions (OR = 4.17, 95 % CI: 1.75 to 9.94, p = 0.001). No significant difference was observed in incidences of biliary stricture, portal vein complications, post-operative bleeding requiring surgical treatments, primary non-function and re-transplantations. In subgroup analyses, biliary and vascular complications only increased after ex-vivo SLT rather than in-situ SLT, and SLT recipients had more re-transplantations if they matched with WLT recipients in terms of urgent status. The authors concluded that adult right lobe SLT was associated with increased biliary and vascular complications compared with WLT, but it did not show significant inferiority in patient and graft survivals.

Ursodeoxycholic Acid for Prevention of Acute Cellular Rejection after Liver Transplantation

Deng et al (2014) noted that acute cellular rejection (ACR) after LT is one of the most common problems faced by transplant recipients in spite of advances in immunosuppressive therapy. Recently, clinical trials reported that ursodeoxycholic acid (UDCA) reduced the incidence of ACR significantly. However, others have shown contradictory conclusion. Therefore, these investigators performed a meta-analysis of rigorous randomized controlled trials (RCTs) to determine the effectiveness of UDCA in reducing ACR after LT. All RCTs that evaluated effectiveness of UDCA as an adjuvant treatment to prevent ACR after LT were searched from PubMed/MEDLINE, EMBASE, Cochrane Central Register of Controlled Trials, ScienceDirect databases and Web of Science (from January 1981 to March 2012). There was no language limitation in these searches. Relevant abstracts of international meetings were also searched. References of each included study were searched manually. A total of 234 patients from 4 high-quality RCTs (Jadad score 4 to 5) were included in this meta-analysis. Prophylactic use of UDCA did not decrease the incidence of ACR (RR: 0.94, 95 % CI: 0.77 to 1.16, p > 0.05), steroid-resistant rejection (RR: 0.77, 95 % CI: 0.47 to 1.27, p > 0.05) and

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the number of patients with the multiple episodes of ACR (RR: 0.60, 95 % CI: 0.28 to 1.30, $p > 0.05$). Different intervention programs (high-dose versus low-dose UDCA; early versus delayed UDCA treatment) also did not alter the outcomes. The authors concluded that UDCA, as an adjuvant treatment, was not able to prevent ACR and steroid-resistant rejection after LT. They stated that further trials should be done to determine whether higher dose of UDCA will be beneficial.

An UpToDate review on "Long-term management of adult liver transplant recipients" (Gaglio and Cotler, 2015) does not mention the use of ursodeoxycholic acid as a management agent.

Xenotransplantation

The success of transplantation has led to a marked increase in the number of candidates to over 16,000 places on the national waiting list. However, there has been little growth in the supply of available cadaveric organs, resulting in an organ shortage crisis. With waiting times often exceeding 1 to 2 years, the waiting list death rate now exceeds 10 % in most regions. Researchers have investigated novel approaches such as xenotransplantation, hepato-cellular transplantation and bioartificial liver to address the growing disparity between the limited supply and excessive demand for suitable organs. However, all these approaches are considered investigational in nature at this juncture.

Studies on xenotransplantation are performed using primates (e.g., baboons, and smaller monkeys). Transmission of diseases, which can be transmitted from animals to humans under natural conditions (zoonoses) as well as hyper-acute rejection remains major concerns in xenotransplantation. Hepatocellular transplantation is used either to temporarily or permanently replace the diseased liver. Hepatocytes are seeded onto biodegradable polymer that serves as a temporary extra-cellular matrix and to induce vascular in-growth. The seeded polymer is then implanted into a vascular rich area, such as the mesentery of the small intestine. Other techniques including direct injection into the spleen or liver. A bioartificial liver is designed to treat liver disease in the manner similar to a dialysis machine treats renal disease. Investigators use porcine hepatocytes or a transformed line of hepatocytes housed in a bioreactor allowing plasma from patients with liver failure to perfuse through it. It can be used either as a bridge to liver transplantation or to allow recovery of the native liver.

Everolimus for Prevention of Organ Rejection

National Institute for Health and Care Excellence (NICE)'s clinical practice guideline on "Everolimus for preventing organ rejection in liver transplantation" (2015) stated that "Everolimus is not recommended within its marketing authorisation for preventing organ rejection in people having a liver transplant".

An UpToDate review on "Liver transplantation in adults: Long-term management of transplant recipients" (Gaglio and Cotler, 2016) states that "Everolimus may be another alternative, though its role in the management of patients following liver transplantation has yet to be established". Furthermore, UpToDate reviews on "Treatment of acute cellular rejection in liver transplantation" (Cotler, 2016) and "" () do not mention everolimus as a management option.

Peri-Operative Immuno-Nutrition

Lei and colleagues (2015) stated that no consensus has been reached concerning the effects of peri-operative immuno-nutrition in patients undergoing liver transplantation. These researchers conducted a meta-analysis to evaluate the effects of peri-operative immuno-nutrition on clinical outcomes and liver function in patients undergoing liver transplantation. The PubMed, Embase, Cochrane Central Register of Controlled Trials, Web of Science, and google scholar were searched to identify all available RCTs that compared peri-operative immuno-nutrition support (arginine, glutamine, ribonucleic acids, and ω-3 polyunsaturated fatty acids) with standard nutrition. The data analysis was performed using Revman 5.2 software. A total of 7 RCTs involving 501 patients were included. Peri-operative immuno-nutrition significantly reduced the risk of infectious complications (RR: 0.51; 95 % CI: 0.27 to 0.98, $p = 0.04$) and shortened the post-operative hospital stay [weighted mean difference (WMD): -3.89; 95 % CI: -7.42 to -0.36; $p = 0.03$]. Furthermore,

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peri-operative immuno-nutrition improved liver function by decreasing the levels of aspartate aminotransferase (AST) in the blood (WMD: -25.4; 95 % CI: -39.9 to -10.9, p = 0.0006). However, these investigators did not find statistically significant differences in serum alanine aminotransferase (ALT), total bilirubin (TB) and direct bilirubin (DB) levels. There were no statistically significant differences in mortality and rejection reaction. The authors concluded that peri-operative nutrition support adding immuno-nutrients like arginine, glutamine, ribonucleic acids, and ω-3 polyunsaturated fatty acids may improve outcomes in patients undergoing liver transplantation. Moreover, they stated that due to the limited sample size of the included trials, further large-scale and rigorously designed RCTs are needed to confirm these preliminary findings.

Peri-Operative Use of Vasopressin

In a meta-analysis, Won and associates (2015) evaluated the effect of peri-operative terlipressin (an analog of vasopressin) on post-operative renal function in patients who have undergone living donor liver transplantation (LDLT) and analyzed the hemodynamic data during transplantation surgery. These investigators assessed the post-operative peak serum creatinine level and changes in the hemodynamic data (e.g., the mean arterial pressure, heart rate, and systemic vascular resistance). They collected RCTs from PubMed, Embase Drugs and Pharmacology, Cochrane Controlled Trials Register, and Cochrane Database on Systematic Reviews. Analysis was conducted using RevMan 5.2. Data from each trial were pooled and weighted by their mean differences and corresponding 95 % CI. A heterogeneity assessment was performed. A total of 3 trials (151 patients) were included. The difference in the mean (95 % CI) peak serum creatinine (mg/dL) levels post-operatively was not significant between the intervention and control groups (WMD: -0.27; CI: -0.55 to 0.01; p = 0.06). Terlipressin significantly decreased heart rate during the anhepatic phase (WMD: -6.58; 95 % CI: -8.85 to -4.31; p < 0.00001) with a low heterogeneity (I(2) = 41 %) and significantly decreased heart rate during the neohepatic phase (WMD: -9.82; 95 % CI: -11.96 to -7.68; p < 0.00001), although the heterogeneity was high (I(2) > 50 %). The authors concluded that an iv infusion of terlipressin peri-operatively for LDLT had no effect on the creatinine values post-operatively. Moreover, they stated that larger RCTs on terlipressin infusions during liver transplantation are needed.

Molecular Adsorbent Recirculating System (MARS)

Khuroo and colleagues (2004) stated that molecular adsorbent recirculating system (MARS), a non-cell based device, is an important option for patients with liver failure to give them additional time for recovery or to serve as a "bridge" to transplantation. However, its effect on survival for such patients is not well known. These researchers evaluated the treatment effects of MARS on patients with acute and acute-on-chronic liver failure. The outcomes measure evaluated was survival. They searched Medline (1966 to 2002) and Embase (1974 to 2002) using the terms liver failure, liver support systems, and MARS. The search was extended to the Cochrane Controlled Trials Registry Database, published abstracts from 5 international conferences, Teraklin (the manufacturer of MARS), known contacts, and bibliographies from each full-published report. They included trials published in English and non-English languages. Eligible studies were randomized and non-randomized controlled trials, which compared the treatment effects of MARS with standard medical treatment. Of the 206 articles screened, 4 randomized controlled trials (RCTs) including 67 patients were analyzed; 2 non-randomized trials with 61 patients were used for explorative analysis. The methodology, population, intervention, and outcomes of each selected trial were evaluated by duplicate independent review. Disagreements were resolved by consensus. In the primary meta-analysis, MARS treatment did not appear to reduce mortality significantly compared with standard medical treatment [relative risk (RR), 0.56; 95 % confidence interval (CI): 0.28 to 1.14; p = 0.11]. Only 1 of the 4 randomized trials analyzed showed significant reduction in mortality. Sensitivity analysis of 3 peer-reviewed trials did not reduce mortality significantly with MARS treatment (RR, 0.72; 95 % CI: 0.37 to 1.40; p = 0.33). Subgroup analysis of 2 trials for acute liver failure and another 2 trails for acute-on-chronic liver failure also did not reveal any benefit to survival with MARS treatment. In contrast, explorative analysis of 2 non-randomized trials showed a significant survival benefit with MARS treatment (risk ratio [RR], 0.36; 95 % CI: 0.17 to 0.76; p = 0.007). This was possibly related to bias in the selection of patients in the non-randomized trials. The authors concluded that MARS treatment had no significant survival benefit on patients with liver failure when compared with standard medical therapy. However, these investigators found only a few trials with a

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small number of patients for the analysis, allowing for the possibility of false negative and erroneous conclusions. They stated that well-conducted randomized trials are strongly recommended to define the role of MARS in the treatment of patients with liver failure.

Vaid and associates (2012) noted that MARS is an artificial liver support system that has been developed for patients with liver failure until the liver regains function or as a bridge to transplantation. These researchers conducted a meta-analysis to examine the effectiveness of this promising therapy. They searched Medline, Embase, and the Cochrane Registry of Controlled Trials databases, and abstracts from the proceedings of several scientific meetings. Patients with acute, acute on chronic, and hyper-acute liver failure were included and these investigators compared MARS with standard medical therapy. Randomized and non-randomized controlled trials were included and Molecular Adsorbent Recirculating System was the intervention used. They evaluated net change in total bilirubin levels, improvement in hepatic encephalopathy and mortality; 9 RCTs and 1 non-randomized controlled study met criteria and were included. By meta-analysis, MARS resulted in a significant decrease in total bilirubin levels (net change -7.0 mg/dL; 95 % CI: -10.4 to -3.7; $p < 0.001$) and in an improvement in the West-Haven grade of hepatic encephalopathy (odds ratio [OR] 3.0; 95 % CI: 1.9 to 5.0; $p < 0.001$). There was no beneficial effect on mortality (OR 0.91; 95 % CI: 0.64 to 1.31; $p = 0.62$). The limitations of this study included a small sample size, an inability to blind with significant heterogeneity among studies, and variable definitions of liver failure. The authors concluded that the MARS is associated with a significant improvement in total bilirubin levels and hepatic encephalopathy; but has no impact on survival. They stated that large studies are needed to assess the merit of this promising therapy on patient-centered outcomes.

Saliba et al (2013) stated that albumin dialysis with the MARS (Gambro, Lund, Sweden), a non-cell artificial liver support device, may be beneficial in acute liver failure (ALF). In a RCT, these investigators examined if MARS improved survival in ALF. Subjects received conventional treatment (n = 49) or MARS with conventional treatment (n = 53), stratified according to whether paracetamol caused ALF. Outcome measures included 6-month survival and secondary end-points included adverse events. A total of 102 patients (mean age of 40.4 years [SD, 13]) were in the modified intention-to-treat (mITT) population. The per-protocol analysis (49 conventional, 39 MARS) included patients with at least 1 session of MARS of 5 hours or more. Six-month survival was 75.5 % (95 % CI: 60.8 % to 86.2 %) with conventional treatment and 84.9 % (CI: 71.9 % to 92.8 %) with MARS ($p = 0.28$) in the mITT population and 75.5 % (C:, 60.8 % to 86.2 %) with conventional treatment and 82.9 % (CI: 65.9 % to 91.9 %) with MARS ($p = 0.50$) in the per-protocol population. In patients with paracetamol-related ALF, the 6-month survival rate was 68.4 % (CI: 43.5 % to 86.4 %) with conventional treatment and 85.0 % (CI: 61.1 % to 96.0 %) with MARS ($p = 0.46$) in the mITT population; 66 of 102 patients had transplantation (41.0 % among paracetamol-induced ALF; 79.4 % among non-paracetamol-induced ALF) ($p < 0.001$). Adverse events did not significantly differ between groups. The authors concluded that this randomized trial of MARS in patients with ALF was unable to provide definitive safety or effectiveness conclusions because many patients had transplantation before administration of the intervention; ALF not caused by paracetamol was associated with greater 6-month patient survival.

He and co-workers (2015) evaluated the treatment effects of the MARS in patients with ALF and acute-on-chronic liver failure (AOCLF). They searched Medline, Embase, and the Cochrane Controlled Trials Registry database between January 1966 and January 2014. They included RCTs, which compared the treatment effects of MARS with standard medical treatment. Study quality assessed according to Consolidated Standards of Reporting Trials (CONSORT) criteria. The RR was used as the effect-size measure according to a fixed-effects model. The search strategy revealed 72 clinical studies, 10 of which were RCTs that met the criteria and were included; 4 addressed ALF (93 patients) and 6 addressed AOCLF (453 patients). The mean CONSORT score was 15 (range of 10 to 20). By meta-analysis, MARS significantly improved survival in ALF (RR 0.61; 95 % CI: 0.38 to 0.97; $p = 0.04$). There was no significant survival benefit in AOCLF (RR 0.88; 95 % CI: 0.74, 1.06; $p = 0.16$). MARS significantly improved survival in patients with ALF, however, there is no evidence that it improved survival in patients with AOCLF. The authors concluded that the present meta-analysis indicated that MARS therapy can improve survival in patients with ALF.

Tsipotis and colleagues (2015) stated that albumin dialysis is the best-studied extra-corporeal non-biologic liver support system as a bridge or destination therapy for patients with liver failure awaiting liver

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transplantation or recovery of liver function. These researchers performed a systematic review to examine the safety and effectiveness of 3 albumin dialysis systems (MARS, fractionated plasma separation, adsorption and hemodialysis [Prometheus system], and single-pass albumin dialysis) in randomized trials for supportive treatment of liver failure. PubMed, Ovid, EMBASE, Cochrane's Library, and ClinicalTrials.gov were searched. Two authors independently screened citations and extracted data on patient characteristics, quality of reports, efficacy, and safety end-points. A total of 10 trials (7 of MARS and 3 of Prometheus) were identified (620 patients). By meta-analysis, albumin dialysis achieved a net decrease in serum total bilirubin level relative to standard medical therapy of 8.0 mg/dL (95 % CI: -10.6 to -5.4) but not in serum ammonia or bile acids. Albumin dialysis achieved an improvement in hepatic encephalopathy relative to standard medical therapy with a RR of 1.55 (95 % CI: 1.16 to 2.08) but had no effect survival with a RR of 0.95 (95 % CI: 0.84 to 1.07). Because of inconsistency in the reporting of adverse events, the safety analysis was limited but did not demonstrate major safety concerns. The authors concluded that the use of albumin dialysis as supportive treatment for liver failure is successful at removing albumin-bound molecules, such as bilirubin and at improving hepatic encephalopathy. They stated that additional experience is needed to guide its optimal use and address safety concerns.

In a prospective, randomized, cross-over study, Sponholz et al (2016) compared 2 devices (MARS and single-pass albumin dialysis (SPAD)) with particular focus on reduction of bilirubin levels (primary end-point) and influence on para-clinical and clinical parameters (secondary end-points) associated with liver failure. Patients presenting with liver failure were screened for eligibility and after inclusion were randomly assigned to be started on either conventional MARS or SPAD (with 4 % albumin and a dialysis flow rate of 700 ml/h). Statistical analyses were based on a linear mixed-effects model. A total of 69 cross-over cycles of extra-corporeal albumin dialysis (ECAD) in 32 patients were completed. Both systems significantly reduced plasma bilirubin levels to a similar extent (MARS: median -68 µmol/L, interquartile range [IQR] -107.5 to -33.5, $p = 0.001$; SPAD: -59 µmol/L, -84.5 to +36.5, $p = 0.001$). However, bile acids (MARS: -39 µmol/L, -105.6 to -8.3, $p < 0.001$; SPAD: -9 µmol/L, -36.9 to +11.4, $p = 0.131$), creatinine (MARS: -24 µmol/L, -46.5 to -8.0, $p < 0.001$; SPAD: -2 µmol/L, -9.0 to +7.0/L, $p = 0.314$) and urea (MARS: -0.9 mmol/L, -1.93 to -0.10, $p = 0.024$; SPAD: -0.1 mmol/L, -1.0 to +0.68, $p = 0.523$) were reduced and albumin-binding capacity was increased (MARS: +10 %, -0.8 to +20.9 %, $p < 0.001$; SPAD: +7 %, -7.5 to +15.5 %, $p = 0.137$) only by MARS. Cytokine levels of interleukin (IL)-6 and IL-8 and hepatic encephalopathy were altered by neither MARS nor SPAD. The authors concluded that both procedures were safe for temporary extra-corporeal liver support. While in clinical practice routinely assessed plasma bilirubin levels were reduced by both systems, only MARS affected other para-clinical parameters (i.e., serum bile acids, albumin-binding capacity, and creatinine and urea levels).

Soo and associates (2016) noted that in children ALF is a rare but life-threatening condition from which 2/3 do not recover with supportive therapy. Treatment is limited by the availability of liver transplants. Molecular adsorbent recirculating system dialysis is a bridge to transplantation that enhances the chances of survival during the waiting period for a transplant, although it cannot improve survival. Open albumin dialysis (OPAL) is a new mode of albumin dialysis developed to further improve dialysis efficiency. These investigators reported a pediatric case of AOCLF and compared the 2 modes of albumin dialysis, namely, the MARS and OPAL, used to treat this patient's cholestatic pruritus. Removal of total and direct bilirubin, ammonia and bile acids were measured by serial blood tests. There was an increased removal of bile acids with the OPAL mode, whereas the removal of total and direct bilirubin and ammonia was similar in both modes. The patient reported better improvement in pruritus following OPAL compared to dialysis with the MARS. The authors concluded that the OPAL may offer a better solution than the MARS in the treatment of refractory pruritus in liver failure.

An UpToDate review on “Acute liver failure in adults: Management and prognosis” (Goldberg and Chopra, 2017) states that “Artificial hepatic assist devices and hepatocyte transplantation -- Attempts have been made to develop an artificial hepatic assist device for acute liver failure that would operate on the same basic principles as hemodialysis for renal failure. However, developing a machine that performs the functions of the liver is inherently more difficult than developing one that performs the excretory functions of the kidneys because the liver performs a large number of diverse and vital synthetic functions. Results in patients treated with these systems have largely been disappointing and the systems are not widely available so they are generally not used in the management of patients with acute liver failure. Support systems designed to treat

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patients with liver failure fall into two main categories, non-cell-based systems, including plasmapheresis, plasma exchange, albumin dialysis, and charcoal-based hemadsorption, and systems that incorporate living hepatocytes or hepatic tissue, also known as bioartificial liver support systems".

Also, there is a phase II clinical trial on "Molecular Adsorbent Recirculating System (MARS®) in Hypoxic Hepatitis (MARS in HH)"; this study is currently recruiting participants. (Last verified September 2016).

Best et al (2019) noted that hepatorenal syndrome is defined as renal failure in people with cirrhosis in the absence of other causes. In addition to supportive treatment such as albumin to restore fluid balance, the other potential treatments include systemic vasoconstrictor drugs (such as vasopressin analogs or noradrenaline), renal vasodilator drugs (such as dopamine), trans-jugular intrahepatic portosystemic shunt (TIPS), and liver support with molecular adsorbent recirculating system (MARS). There is uncertainty over the best treatment regimen for hepatorenal syndrome. In a Cochrane review, these investigators compared the benefits and harms of different treatments for hepatorenal syndrome in people with decompensated liver cirrhosis. The authors concluded that based on very low-certainty evidence, there is no evidence of benefit or harm of any of the interventions for hepatorenal syndrome with regards to the following outcomes: all-cause mortality, serious adverse events (proportion), number of serious adverse events per participant, any adverse events (proportion), liver transplantation, or other decompensation events. Low-certainty evidence suggested that albumin plus noradrenaline had fewer 'any adverse events per participant' than albumin plus terlipressin. Low- or very low-certainty evidence also found that albumin plus midodrine plus octreotide and albumin alone had lower recovery from hepatorenal syndrome compared with albumin plus terlipressin. These researchers stated that future randomized clinical trials should be adequately powered; employ blinding, avoid post-randomization drop-outs or planned cross-overs (or perform an intention-to-treat [ITT] analysis); and report clinically important outcomes such as mortality, health-related quality of life (HR-QOL), adverse events, and recovery from hepatorenal syndrome. These investigators stated that albumin plus noradrenaline and albumin plus terlipressin appeared to be the interventions that should be compared in future trials.

Kade et al (2020) examined the effectiveness of MARS in patients with alcohol-related acute-on-chronic liver failure (AoCLF) complicated with type 1 hepatorenal syndrome (HRS). So far, MARS efficacy and safety has been demonstrated in various acute liver failure scenarios. Data from 41 MARS procedures (10 patients with type 1 HRS, in the course of alcohol-related AoCLF were considered for this study. Biochemical tests of blood serum were performed before and after each procedure. The condition of patients was determined before and after the treatment with the use of the model for end-stage liver disease - sodium (MELD-Na) and the stage of encephalopathy severity based on the West Haven criteria. During the observation period (20.5 ± 13.9 days), 5 patients died, and the remaining 5 surviving patients were discharged from the hospital. In the group of 10, the 14-day survival, starting from the 1st MARS treatment, was 90 %. The MARS procedure was associated with a 19 % reduction in bilirubin (27.5 ± 6.1 versus 22.3 ± 4.0 mg/dL, $p < 0.001$), 37 % reduction in ammonia (44.1 ± 22.5 versus 27.6 ± 20.9 $p < 0.001$), 27 % reduction in creatinine (1.5 ± 1.0 versus 1.1 ± 0.6 mg/dL, $p < 0.001$) and 14 % reduction urea (83.8 ± 36.1 versus 72.1 ± 33.3, $p < 0.001$) in blood serum samples, with stable hemodynamic parameters. In the group of patients discharged from the clinic ($n = 5$), the MARS treatments resulted in an improvement in hepatic encephalopathy (West Haven; $p = 0.043$), as well as a reduction in the MELD-Na score ($p = 0.015$). The authors concluded that MARS is a hemodynamically safe method for supporting the function of the liver and the kidneys. Application of the MARS reduced the symptoms of encephalopathy in patients with alcohol-related type 1 HRS. Moreover, these researchers stated that despite promising observations, the findings of this study were subject to limitations caused primarily by the small number of patients resulting from the selection of a homogeneous group of patients.

Sparrelid et al (2020) stated that post-hepatectomy liver failure (PHLF) remains a serious complication after major liver resection with severe 90-day mortality; MARS is a potential therapeutic option in PHLF. In a systematic review, these investigators examined the experiences and results of MARS in PHLF. Following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, they carried out a systematic literature review using PubMed and Embase; non-randomized trials were assessed by the MINORS criteria. A total of 2,884 records were screened and 22 studies were extracted (no RCT). They contained 809 patients including 82 patients with PHLF; 5 studies ($n = 34$) specifically examined the role of

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MARS in patients with PHLF. In these patients, overall 90-day survival was 47 %. Patients with primary PHLF had significantly better 90-day survival compared to patients with secondary PHLF (60 % versus 14 %, p = 0.03) and treatment was started earlier (median POD 6 (range of 2 to 21) versus median POD 30 (range of 15 to 39); p < 0.001). Number of treatments differed non-significantly in these groups. Safety and feasibility of early MARS treatment following hepatectomy was demonstrated in 1 prospective study; no major adverse events (AEs) have been reported. The authors concluded that early MARS treatment was safe and feasible in patients with PHLF. Currently, MARS cannot be recommended as standard of care in these patients; further prospective studies are needed.

Soreide and Deshpande (2021) stated that PHLF is a relatively rare but feared complication following liver surgery, and is associated with high morbidity, mortality and cost implications. Significant advances have been made in detailed pre-operative assessment, particularly of the liver function in an attempt to predict and mitigate this complication. These researchers carried out a detailed search of PubMed and Medline using keywords "liver failure", "liver insufficiency", "liver resection", "postoperative", and "post-hepatectomy". Only full texts published in English were considered. Particular emphasis was placed on literature published after 2015. A formal systematic review was not found feasible hence a pragmatic review was performed. The reported incidence of PHLF varies widely in reported literature due to a historical absence of a universal definition. Incorporation of the now accepted definition and grading of PHLF would suggest the incidence to be between 8 % and 12 %. Major risk factors include background liver disease, extent of resection and intra-operative course. The vast majority of mortality associated with PHLF was related to sepsis, organ failure and cerebral events. Despite multiple attempts, there has been little progress in the definitive and specific management of liver failure. These investigators discussed recent advances made in detailed pre-operative evaluation of liver function and evidence-based targeted approach to managing PHLF. The authors concluded that PHLF remains a major cause of mortality following liver resection. In absence of a specific remedy, the best approach is mitigating the risk of it happening by detailed assessment of liver function, patient selection and general care of a critically ill patient. Artificial liver support was one of the keywords listed of this review.

Furthermore, the current version of UpToDate review on “Acute liver failure in adults: Management and prognosis” (Goldberg et al, 2021) states that “Artificial hepatic assist devices -- Attempts have been made to develop an artificial hepatic assist device for acute liver failure that would operate on the same basic principles as hemodialysis for renal failure. However, developing a machine that performs the functions of the liver is inherently more difficult than developing one that performs the excretory functions of the kidneys because the liver performs a large number of diverse and vital synthetic functions. Results in patients treated with these systems have largely been disappointing and the systems are not widely available. As a result, they are generally not used in the management of patients with acute liver failure”.

Normothermic Machine Perfusion for Liver Transplantation

Bral and co-workers (2017) stated that after extensive experimentation, outcomes of a first clinical normothermic machine perfusion of the liver (NMP-L) trial in the United Kingdom demonstrated feasibility and clear safety, with improved liver function compared with standard static cold storage (SCS). These researchers presented a preliminary single-center North American experience using identical NMP technology. Total of 10 donor liver grafts were procured, 4 (40 %) from donation after circulatory death (DCD), of which 9 were transplanted. One liver did not proceed because of a technical failure with portal cannulation and was discarded. Transplanted NMP grafts were matched 1:3 with transplanted SCS livers. Median NMP was 11.5 hours (range of 3.3 to 22.5 hours) with 1 DCD liver perfused for 22.5 hours. All transplanted livers functioned, and serum transaminases, bilirubin, international normalized ratio, and lactate levels corrected in NMP recipients similarly to controls. Graft survival at 30 days (primary outcome) was not statistically different between groups on an intent-to-treat basis (p = 0.25). Intensive care and hospital stays were significantly more prolonged in the NMP group. The authors concluded that this preliminary experience demonstrated feasibility as well as potential technical risks of NMP in a North American setting and highlighted a need for larger, randomized studies.

Laing and associates (2017) noted that NMP-L is a novel technology recently introduced into the practice of liver transplantation. These researchers discussed benefits of normothermic perfusion over conventional

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SCS and summarized recent publications in this area. The first clinical trials have demonstrated both safety and feasibility of NMP-L. They have shown that machine perfusion can entirely replace cold storage or be commenced following a period of cold ischemia. The technology currently allows transplant teams to extend the period of organ preservation for up to 24 hours. Results from the first RCT comparing NMP-L with SCS will be available soon. One major advantage of NMP-L technology over other parallel technologies is the potential to assess liver function during NMP-L. Several case series have suggested parameters usable for liver viability testing during NMP-L including bile production and clearance of lactic acidosis. NMP-L allows viability testing of high-risk livers. It has shown the potential to increase utilization of donor organs and improve transplant procedure logistics. The authors concluded that NMP-L is likely to become an important technology that will improve organ preservation as well as have the potential to improve utilization of extended criteria donor livers.

Ceresa and colleagues (2017) stated that preservation of the liver via NMP is rapidly becoming an area of great academic and clinical interest. These investigators described the benefits and limitations of NMP and where the role for SCS may lie. Clinical studies have recently been published reporting the use of NMP in liver preservation for transplantation. They have described the technology to be well-tolerated and feasible with potentially improved post-transplant outcomes; NMP facilitates extended preservation times as well as the potential to increase organ utilization through viability assessment and regeneration. However, this technology is considerably more costly than cold storage and carries significant logistical challenges. Cold storage remains the gold standard preservation for standard criteria livers with good long-term patient and graft survival. The authors concluded that NMP is an exciting new technological advancement in liver preservation, which is likely to have a positive impact in liver transplantation. However, they stated that RCTs are needed to justify its inclusion into standard practice and provide evidence to support its effectiveness.

Lai et al (2025) noted that LT is the gold standard for ESLD; however, ischemic cholangiopathy (IC) remains a significant complication. Ex-situ normothermic machine perfusion (ES-NMP) has emerged as a potential strategy to mitigate ischemic injury. However, the effect of ESNMP on reducing post-LT IC remains controversial. These investigators carried out an updated meta-analysis to examine the impact of ES-NMP on the incidence of IC. They conducted a systematic review and meta-analysis following PRISMA guidelines. The literature search included studies from 2015 to 2025 comparing LT outcomes using ES-NMP versus SCS. The primary outcome was the incidence of IC. Risk of bias was evaluated using the ROBINS-E tool. Statistical analysis, including random-effects meta-analysis, sensitivity analysis, and meta-regression, was carried out to assess heterogeneity, potential confounders, and the impact of follow-up duration. A total of 17 studies, including 76,045 patients (4,843 ES-NMP; 71,202 SCS), were analyzed. No statistically significant difference in IC incidence was found between ES-NMP and SCS (1.3 % versus 0.6 %; RR = 0.68, 95 % CI: 0.41 to 1.13; p = 0.14). Sensitivity analysis excluding 1 outlier study showed a reduction in IC risk with ES-NMP (RR = 0.62, 95 % CI: 0.38 to 1.01; p = 0.054). Two sub-analyses of studies with 12 months or longer of follow-up (RR = 0.51, 95 % CI: 0.26 to 0.99; p = 0.049) and DCDs (RR = 0.33, 95 % CI: 0.16 to 0.67; p = 0.002) demonstrated risk reduction. The meta-regression showed that the back-to-base perfusion approach was associated with the occurrence of IC, with an OR of 1.03 (95 % CI: 1.00 to 1.07, p = 0.035). The authors concluded that there appeared a correlation between ES-NMP use and IC reduced risk, especially with longer follow-up periods and DCDs, although further high-quality studies are needed to confirm benefits and refine clinical use.

Pre-Operative Neutrophil-to-Lymphocyte Ratio

Xu and co-workers (2018) stated that many recent reports showed that the pre-transplant neutrophil-to-lymphocyte ratio (NLR) may be correlated with the prognosis of patients undergoing liver transplantation for HCC. However, their results still remained controversial. These researchers performed a meta-analysis of 13 studies to estimate the prognostic value of pre-transplant NLR. Databases including PubMed, Embase, Cochrane Library and Web of Science were searched to September 2017. Hazard ratio (HR) or OR with its 95 % CI was used to evaluate the association between elevated NLR and the prognosis or clinical features of liver cancer patients. A total of 13 studies including 1,936 patients were included in this meta-analysis. Elevated pre-transplant NLR had a close association with the overall survival (OS) (HR: 2.22; 95 % CI: 1.34

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to 3.68), recurrence-free survival (RFS) (HR: 3.77; 95 % CI: 2.01 to 7.06) and disease-free survival (DFS) (HR: 2.51; 95 % CI: 1.22 to 5.15) of patients undergoing LT for HCC, respectively. In addition, elevated NLR was associated with the presence of vascular invasion (OR: 2.39; 95 % CI: 1.20 to 4.77) and Milan criteria (OR: 0.26; 95 % CI: 0.17 to 0.40). The authors concluded that the results of this meta-analysis showed that elevated pre-transplant NLR may be used as a new prognostic predictor after liver transplantation for HCC.

Pre-Operative Platelet-to-Lymphocyte Ratio

Lai and colleagues (2018) performed a systematic review and meta-analysis on platelet-to-lymphocyte ratio (PLR) as a risk factor for post-transplant HCC recurrence. A systematic literature search was performed using PubMed. Participants of any age and sex, who underwent liver transplantation for HCC were considered following the following criteria: studies comparing pre-transplant low versus high PLR values; studies reporting post-transplant recurrence rates; and if more than 1 study was reported by the same institute, only the most recent was included. The primary outcome measure was set for HCC recurrence after transplantation. A total of 5 articles, published between 2014 and 2017, fulfilled the selection criteria. As for the quality of the reported studies, all the investigated articles presented an overall high quality. A total of 899 cases were investigated: 718 cases (80.0 %) were males; 3 studies coming from European countries and 1 from Japan presented HCV as the main cause of cirrhosis. On the opposite, 1 Chinese study presented a greater incidence of HBV-related cirrhotic cases. In all the studies apart 1, the PLR cut-off value of 150 was reported. At meta-analysis, high PLR value was associated with a significant increase in recurrence after transplantation (OR = 3.33; 95 %CI: 1.78 to 6.25; $p < 0.001$). A moderate heterogeneity was observed among the identified studies according to the Higgins I2 statistic value. The authors concluded that pre-transplant high PLR values were connected with an increased risk of post-operative recurrence of HCC. Moreover, they stated that although the reported data suggested an effective biological correlation between platelets and tumor aggressive behavior, they underlined that further clinical studies trying to univocally demonstrate the biological role of platelets in the HCC oncogenesis are needed. The authors stated that this study had 2 main drawbacks. First, moderate heterogeneity was observed among the studies investigated, as clearly shown by the reported Higgins I2 statistic value (26.8 %). Such a phenomenon was surely caused by the broad eligibility criteria for HCC and the different PLR cut-off values used in the different centers. It was, in fact, clear that a meta-regression weighted for the geographical area, HCV versus HBV as the main cause of liver failure, living-donor versus deceased-donor LT, and markers of tumor aggressiveness should represent a more accurate way for better clarify the role of PLR in this setting. Unfortunately, the limited number of cases reported did not consent these investigators to perform more sophisticated analyses. Secondly, no information was reported in the different series on the presence and grade of portal hypertension, a very well known cause of thrombocytopenia in cirrhosis.

Transient Elastography in Acute Cellular Rejection Following Liver Transplantation

Bhat and associates (2017) stated that recurrent fibrosis after liver transplantation impacts on long-term graft and patient survival. These investigators performed a meta-analysis to compare the accuracy of non-invasive methods to diagnose significant recurrent fibrosis (stage F2 to F4) following liver transplantation. Studies comparing serum fibrosis biomarkers, namely AST-to-platelet ratio index (APRI), fibrosis score 4 (FIB-4), or transient elastography (TE) with liver biopsy in liver transplantation recipients were systematically identified through electronic databases. In the meta-analysis, these researchers calculated the weighted pooled OR and used a fixed effect model, as there was no significant heterogeneity between studies. A total of 8 studies were included for APRI, 4 for FIB-4, and 12 for TE. The mean prevalence of significant liver fibrosis was 37.4 %. The summary OR was significantly higher for TE (21.17, 95 % CI: 14.10 to 31.77, p = 1X10-30) as compared to APRI (9.02, 95 % CI: 5.79 to 14.07; p = 1X10-30) and FIB-4 (7.08, 95 % CI: 4.00 to 12.55; p = 1.93X10-11). The authors concluded that TE performed best to diagnose recurrent fibrosis in liver transplantation recipients; APRI and FIB-4 can be used as an estimate of significant fibrosis at centers where TE is not available. They stated that longitudinal assessment of fibrosis by means of these non-invasive tests may reduce the need for liver biopsy.

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The authors stated that this study had several drawbacks, including the higher proportion of hepatitis C virus patients represented. These investigators did also include pediatric studies, with various reasons for development of fibrosis post-transplant. Finally, the reported range of cut-off values for TE was wide (from 7.3 kPa to 12.3 kPa) in liver transplantation recipients, rendering potentially complicated the use in clinical practice for the individual patients. This finding was likely due to highly heterogeneous study populations due to etiology of liver disease, different post-LT time points for the assessment of liver fibrosis, study design and sample size and variable interval range between liver biopsy and the non-invasive assessment of liver fibrosis. These researchers could not account for these different cut-offs for TE for the various etiologies of chronic liver diseases in this meta-analysis, as these were not provided in the studies. Nonetheless, the validity of these findings across this heterogeneous population suggested the usefulness of these non-invasive tools for fibrosis even in a complex context such as that of liver transplantation.

Moreover, these researchers noted that given non-invasiveness and feasibility for serial measurements, non-invasive tests for liver fibrosis could be used in the post-transplant clinical setting as an additive tool for suspected recurrent or de-novo liver disease. The high accuracy they found in this meta-analysis, especially for TE, suggested that these tests have similar diagnostic value as in the pre-transplant setting. It should be underlined that liver biopsy remains a cornerstone for the clinical management of liver transplantation recipients, as non-invasive tests cannot differentiate between different liver pathologies that can co-exist in this setting, such as acute or chronic rejection. Nonetheless, when the etiology of recurrent or de-novo disease has been established, these non-invasive tests are helpful in following fibrosis progression longitudinally and implementing preventive therapies in a timely manner. They stated that further studies aimed at defining optimal cut-off values for definition of significant liver fibrosis are needed.

Nacif and colleagues (2018) noted that TE is a non-invasive technique that measures liver stiffness. When an inflammatory process is present, this is shown by elevated levels of stiffness. Acute cellular rejection (ACR) is a consequence of an inflammatory response directed at endothelial and bile epithelial cells, and it is diagnosed through liver biopsy. This is a systematic review of the viability of TE in ACR following liver transplantation. The Cochrane Library, Embase, and Medline PubMed databases were searched and updated to November 2016. The MESH terms used were "liver transplantation", "graft rejection", "elasticity imaging techniques" (PubMed), and "elastography" (Cochrane and Embase). A total of 70 studies were retrieved and selected using the PICO (patient, intervention, comparison or control, outcome) criteria; 3 prospective studies were selected to meta-analysis and evaluation. A total of 33 patients with ACR were assessed with TE. One study showed a cut-off point of greater than 7.9 kPa to define graft damage and less than 5.3 kPa to exclude graft damage (ROC of 0.93; $p < 0.001$). Another study showed elevated levels of liver stiffness in ACR patients. However, in this study, no cut-off point for ACR was suggested. The final prospective study included 27 patients with ACR at liver biopsy. Cut-off points were defined as TE greater than 8.5 kPa, moderate-to-severe ACR, with a specificity of 100 % and ROC of 0.924. The measurement of TE less than 4.2 kPa excluded the possibility of any ACR ($p = 0.02$). The authors concluded that TE may be an important tool for the severity of ACR in patients following liver transplantation. Moreover, they stated that further studies should be performed to better define the cut-off points and applicability of the examination.

Biomarkers for Diagnosis of Acute Allograft Rejection following Liver Transplantation

Krenzien and colleagues (2019) stated that non-invasive blood and urine markers have been widely explored in recent decades for diagnosing acute rejection following liver transplantation. However, none has been translated into routine clinical use so far due to uncertain diagnostic accuracy, and liver biopsy remains the gold standard. These investigators performed a systematic review and meta-analysis of diagnostic biomarkers for non-invasive diagnosis of acute allograft rejection following liver transplantation. Systematic literature searches of Medline, Cochrane and Embase were conducted up to February 2019 to identify studies evaluating the use of non-invasive markers in diagnosing allograft rejection following liver transplantation. Meta-analysis was performed using a random effects model with DerSimonian-Laird weighting and the hierarchical summary receiver operating curve. Of 560 identified studies, 15 studies (1,445 patients) met the inclusion criteria. The following markers were tested: acid labile nitroso-compounds (NOx), serum amyloid A protein, procalcitonin, peripheral blood eosinophil count, peripheral blood T-cell activation and interleukin 2 (IL-2) receptor, guanylate-binding protein-2 mRNA, graft-derived cell-free DNA, pi-glutathione S-

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transferase, alpha-glutathione S-transferase and serum HLA class I soluble antigens. Only eosinophil count was tested in multiple studies, and they demonstrated high heterogeneity (I 2 = 72 % [95 % CI: 0.5 to 0.99]). IL-2 receptor demonstrated the highest sensitivity (89 % [95 % CI: 0.78 to 0.96]) and specificity (81 % [95 % CI: 0.69 to 0.89]). The authors concluded that IL-2 receptor expression demonstrated the highest diagnostic accuracy, while the peripheral eosinophil count was the only marker tested in more than 1 study. These researchers stated that currently, liver biopsy remains superior to non-invasive diagnostic biomarkers as most studies exhibited inferior designs, hindering possible translation into clinical application at this time.

Factor V Leiden and F2 Testing is for Individuals Scheduled to Receive Partial Liver Transplant for Primary Sclerosing Cholangitis

Perez-Pujol and colleagues (2012) noted that a more limited information is available on the impact of factor V Leiden (FVL) in clinical events where inflammation is prevalent. Uremia, cirrhosis, liver transplantation, but also sepsis (generalized inflammatory state also known as SIR-systemic inflammatory response), infection, and inflammatory bowel disease (IBD) result in a chronic inflammatory state. Even though, both mechanisms of action and clinical management of the previous conditions are well established, it has not been possible to reach a final consensus on whether FVL is able to modify the outcome of these events.

Fan and associates (2013) stated that hepatic artery thrombosis (HAT) after orthotopic liver transplantation (OLT) is associated with significant morbidity and mortality; FVL mutation is the most common genetic defect that predisposes to thrombosis. The reconstruction of hepatic artery with arterial graft is a documented risk factor for HAT. However, the relationship among FVL mutation, arterial graft, and HAT remains to be determined. These researchers randomly genotyped 485 patients who underwent OLT from April 2002 to January 2011; and studied the incidence of HAT in the presence of FVL mutation. Of 485 patients, 21 patients (4.3 %) developed HAT (13 men, 8 women); 10 patients (4 men, 6 women) were heterozygous for the FVL mutation. The incidences of HAT in patients without versus with the FVL mutation were 3.8 % and 30 % (p = 0.007). Of patients with HAT, 8 hepatic arteries were reconstructed with infra-renal aortic conduits. All 3 patients (100 %) with versus 5 (28 %) without FVL who received arterial grafts developed HAT (p = 0.042). The authors concluded that the findings of this study suggested that the FVL mutation may be a risk factor for HAT in liver transplantation; the risk was augmented in the presence of an arterial graft.

Kupeli and co-workers (2015) noted that solid-organ transplant recipients can develop chronic hyper-coagulation that increases the incidence of pulmonary embolism (PE). These investigators evaluated the frequency of PE in solid-organ transplant recipients during the first 10 years after transplantation and evaluated the risk factors for its development. The medical records of solid-organ transplant recipients who were treated between 2003 and 2013 were retrospectively reviewed. The reviewed data included demographics, type of transplant, co-morbidities, pro-coagulation factors, thrombo-embolism prophylaxis, and the timing and extent of PE. In total, 999 solid-organ transplant recipients were included in this study (661 renal and 338 liver transplant recipients) (male : female ratio = 665 : 334). 12 renal (1.2 %) and 1 liver transplant recipient (0.3 %) were diagnosed with PE, which developed 1 year after transplantation in 10 patients: 1 patient developed PE less than 3 months after transplantation, and the other 9 patients developed PE within 3 to 6 months. No patients had a prior history of deep venous thrombosis (DVT) or PE; 5 patients received tacrolimus, 7 patients received sirolimus, and 1 patient received cyclosporine; 10 patients received prednisolone, and 8 patients received mycophenolate mofetil. All patients were homozygous normal for FVL and prothrombin genes; 1 patient was homozygous abnormal, and 1 patient had a heterozygous mutation in the methylenetetrahydrofolate reductase (MTFR) gene; 2 patients were treated with low-molecular-weight heparin (LMWH), while the remaining patients received warfarin; 8 patients were treated for 6 months, and the remainder received longer treatments. The authors concluded that the incidence of PE in solid-organ transplant recipients was 1.2 %. Renal transplant recipients were at higher risk of developing PE than liver transplant recipients. The factors that increase the risk of PE in solid-organ transplant recipients appeared to be multi-factorial and included genetic predisposition.

Bagheri Lankarani and colleagues (2015) stated that portal vein thrombosis (PVT) is a fairly common and potentially life-threatening complication in patients with liver cirrhosis. The risk factors for PVT in these

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patients are still not fully understood. These investigators examined the associations between various risk factors in cirrhotic patients and the development of PVT. In this case-control study, these researchers studied 219 patients (greater than 18 years old) with liver cirrhosis, who were awaiting liver transplantation. They were evaluated by history, physical examination, and laboratory tests, including FVL, prothrombin gene mutation, Janus Kinase 2 (JAK2) mutation, and serum levels of protein C, protein S, anti-thrombin III, homocysteine, factor VIII, and anti-cardiolipin antibodies. There was no statistically significant difference in the assessed hyper-coagulable states between patients with or without PVT. A history of previous variceal bleeding with subsequent endoscopic treatment in patients with PVT was significantly higher than in those without it (p = 0.013, OR: 2.526, 95 % CI: 1.200 to 5.317). The author concluded that in this population of cirrhotic patients, treatment of variceal bleeding predisposed the patients to PVT, however, hyper-coagulable disorders by themselves were not associated with PVT.

Furthermore, an UpToDate review on "Liver transplantation in adults: Patient selection and pretransplantation evaluation" (Dove and Brown, 2019) does not mention FVL testing or F2 testing as a management tool.

Neutrophil Gelatinase-Associated Lipocalin (NGAL) in Predicting Acute Kidney Injury Following Orthotopic Liver Transplantation

Yeung and colleagues (2018) noted that acute kidney injury (AKI) is common following orthotopic liver transplantation (OLT) usually occurring early post-transplant. Multiple causes include graft preservation injury, blood loss, hypotension but also severity of recipient liver disease. Early intervention in AKI has both short- and long-term patient benefits. Unfortunately there are no current clinical biomarkers of early AKI. In a systemic review, these investigators examined the value of NGAL in predicting AKI following OLT. Ovid Medline and Embase were searched between the years of 2000 and 2017 for studies using keywords: Neutrophil gelatinase associated lipocalin or NGAL variants combined with synonyms for liver transplantation. A total of 96 studies were identified; 11 studies including 563 patients were considered suitable for analysis. Both urinary (uNGAL) and plasma NGAL (pNGAL) measurement were found to predict AKI following liver transplantation. Optimal reported area under the receiver-operator characteristics curve (AUROC) values of 0.5 to 0.83 and 0.54 to 0.86 respectively. The authors concluded that NGAL is a good predictor of early AKI following OLT although there was considerable variation in the published results. These researchers stated that further studies with prospectively defined cut-off values, standardized definitions of AKI and rigorous data reporting should be conducted to establish its clinical usefulness and limitations.

Basiliximab for Induction Immunosuppression in Individuals Undergoing Liver Transplantation

Best and colleagues (2020) stated that LT is considered the definitive therapy for individuals with liver failure. As part of post-LT management, immunosuppression is given to prevent graft rejections. Immunosuppressive drugs can be classified into those that are used for a short period during the beginning phase of immunosuppression (induction immunosuppression) and those that are used over the entire lifetime of the individual (maintenance immunosuppression), because it is widely believed that graft rejections are more common during the first few months after LT. Some drugs such as glucocorticoids may be used for both induction and maintenance immunosuppression because of their multiple modalities of action. There is considerable uncertainty as to whether induction immunosuppression is necessary and if so, the relative efficacy of different immunosuppressive agents. These researchers examined the comparative benefits and harms of different induction immunosuppressive regimens in adults undergoing LT through a network meta-analysis and generated rankings of the different induction immunosuppressive regimens according to their safety and efficacy. They searched CENTRAL, Medline, Embase, Science Citation Index Expanded, World Health Organization International Clinical Trials Registry Platform, and trials registers until July 2019 to identify randomized clinical trials in adults undergoing LT. These investigators included only randomized clinical trials (irrespective of language, blinding, or status) in adults undergoing LT. They excluded randomized clinical trials in which subjects had multi-visceral transplantation and those who already had graft rejections. They performed a network meta-analysis with OpenBUGS using Bayesian methods and

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calculated the OR, rate ratio, and HR with 95 % CIs based on an available case analysis, according to National Institute of Health and Care Excellence Decision Support Unit guidance. These researchers included a total of 25 trials (3,271 subjects; 8 treatments) in the review; 23 trials (3,017 subjects) were included in one or more outcomes in the review. The trials that provided the information included individuals undergoing primary LT for various indications and excluded those with HIV and those with renal impairment. The follow-up in the trials ranged from 3 to 76 months, with a median follow-up of 12 months among trials. All except 1 trial were at high risk of bias, and the overall certainty of evidence was very low. Overall, approximately 7.4 % of individuals who received the standard regimen of glucocorticoid induction died and 12.2 % developed graft failure. All-cause mortality and graft failure was lower with basiliximab compared with glucocorticoid induction: all-cause mortality (HR 0.53, 95 % CI: 0.31 to 0.93; network estimate, based on 2 direct comparison trials (131 subjects; low-certainty evidence)); and graft failure (HR 0.44, 95 % CI: 0.28 to 0.70; direct estimate, based on 1 trial (47 subjects; low-certainty evidence)). There was no evidence of differences in all-cause mortality and graft failure between other induction immunosuppressants and glucocorticoids in either the direct comparison or the network meta-analysis (very low-certainty evidence). There was also no evidence of differences in serious adverse events (AEs; proportion), serious AEs (number), renal failure, any AEs (proportion), any AEs (number), liver re-transplantation, graft rejections (any), or graft rejections (requiring treatment) between other induction immunosuppressants and glucocorticoids in either the direct comparison or the network meta-analysis (very low-certainty evidence). However, because of the wide CIs, clinically important differences in these outcomes cannot be ruled out. None of the studies reported health-related quality of life (QOL). The authors concluded that based on low-certainty evidence, basiliximab induction may decrease mortality and graft failure compared to glucocorticoids induction in individuals undergoing LT. However, there is considerable uncertainty regarding this finding because this information was based on small trials at high risk of bias. The evidence was uncertain regarding the effects of different induction immunosuppressants on other clinical outcomes, including graft rejections. These researchers stated that future randomized clinical trials should be adequately powered, employ blinding, avoid post-randomization drop-outs (or perform ITT analysis), and use clinically important outcomes such as mortality, graft failure, and health-related QOL. It should be noted that the source of funding for 14 trials was drug companies who would benefit from the results of the study; 2 trials were funded by neutral organizations who had no vested interests in the results of the study; and the source of funding for the remaining 9 trials was unclear.

Hypothermic Machine Perfusion for Reduction of the Incidences of Early Allograft Dysfunction and Biliary Complications after Liver Transplantation

Bellini and associates (2019) noted that to match the current organ demand with organ availability from the donor pool, there has been a shift towards acceptance of extended criteria donors (ECD), often associated with longer ischemic times. Novel dynamic preservation techniques as hypothermic or normothermic machine perfusion (HMP or NMP) are increasingly adopted, especially for organs from ECDs. In a systematic review, these researchers compared the viability and incidence of reperfusion injury in kidneys and livers preserved with MP versus static cold storage (SCS). They carried out a systematic review and meta-analysis between February and March 2019. Medline, Embase and Transplant Library were searched via OvidSP. The Cochrane Library and The Cochrane Central Register of Controlled Trials (CENTRAL) were also searched. English language filter was applied. The systematic search generated 10,585 studies, finally leading to a total of 30 studies for meta-analysis of kidneys and livers; HMP statistically significantly lowered the incidence of primary non-function (PNF, $p = 0.003$) and delayed graft function (DGF, $p < 0.00001$) in kidneys compared to SCS, but not its duration. No difference was also noted for serum creatinine or estimated glomerular filtration rate (eGFR) post-transplantation, but overall kidneys preserved with HMP had a significantly longer 1-year graft survival (OR: 1.61 95 % CI: 1.02 to 2.53, $p = 0.04$). Differently from kidneys where the graft survival was affected, there was no significant difference in PNF for livers stored using SCS for those preserved by HMP and NMP. Machine perfusion demonstrated superior outcomes in early allograft dysfunction (EAD) and post-transplantation AST levels compared to SCS, but however, only HMP was able to significantly decrease serum bilirubin and biliary stricture incidence compared to SCS. The authors concluded that MP improved DGF and 1-year graft survival in kidney transplantation; it appeared to mitigate EAD in livers; however, more clinical studies are needed for

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verification with homogeneous parameters to measure the outcomes of interest, and to prove the potential superiority of these novel technologies in relation to PNF in livers.

Zhang and colleagues (2019) stated that the worldwide organ shortage continues to be the main limitation of LT. To bridge the gap between the demand and supply of liver grafts, it becomes necessary to use extended criteria for donor livers. Hypothermic machine perfusion is designed to improve the quality of preserved organs before implantation. In clinical LT, HMP is still in its infancy. These researchers carried out a systematic search of the PubMed, Embase, Springer, and Cochrane Library databases to identify studies comparing the outcomes in patients with HMP versus SCS of liver grafts. The parameters analyzed included the incidences of PNF, EAD, vascular complications, biliary complications, length of hospital stay (LOS), and 1-year graft survival. A total of 6 studies qualified for the review, involving 144 and 178 liver grafts with HMP or SCS preservation, respectively. The incidences of EAD and biliary complications were significantly reduced with an OR of 0.36 (95 % CI: 0.17 to 0.77, p=0.008) and 0.47 (95 % CI: 0.28 to 0.76, p=0.003), respectively, and 1-year graft survival was significantly increased with an OR of 2.19 (95 % CI: 1.14 to 4.20, p=0.02) in HMP preservation compared to SCS. However, there was no difference in the incidence of PNF (OR 0.30, 95 % CI: 0.06 to 1.47, p=0.14), vascular complications (OR 0.69, 95 % CI: 0.29 to 1.66, p=0.41), and the LOS (MD -0.30, 95 % CI: -4.10 to 3.50, p =0.88) between HMP and SCS preservation. The authors concluded that HMP was associated with a reduced incidence of EAD and biliary complications, as well as an increased 1-year graft survival; however, it was not associated with the incidence of PNF, vascular complications, and the LOS. Moreover, these researchers stated that d due to the drawbacks of this meta-analysis, further large, multi-center RCTs are needed to confirm these findings.

The authors stated that there were several potential drawbacks to this meta-analysis, which may have increased the possibility of publication bias and affected the final result. First, this meta-analysis contained only 6 studies and the number of cases was still limited for this special subject. Second, all of them were cohort studies. Although they provided the best evidence available on this subject, the non-randomized studies might have resulted in an unbalanced selection of patients. Third, there was heterogeneity in the graft quality or donor status, including the length of warm ischemic time and cold ischemic time, donor age, steatosis, types of machine perfusion solution, perfusion route, perfusion pressure, and with or without active oxygenation, which was correlated with the study design and the preferences of individual hospitals.

Patrono et al (2022) noted that prolonged warm ischemia time (WIT) has a negative prognostic value in LT using grafts procured after DCD. These investigators examined the value of abdominal normothermic regional perfusion (A-NRP) associated with dual hypothermic oxygenated machine perfusion (D-HOPE) in controlled DCD LT. They prospectively analyzed data on LTs performed between January 2016 and July 2021. Outcome of controlled DCD LTs performed using A-NRP + D-HOPE (n = 20) were compared to those performed with grafts procured after DBD (n = 40), selected using propensity-score matching. DCD utilization rate was 59.5 %. In the DCD group, median functional WIT, A-NRP and D-HOPE time was 43, 246, and 205 mins, respectively. Early outcomes of DCD grafts recipients were comparable to those of matched DBD LTs. In DCD and DBD group, incidence of anastomotic biliary complications and ischemic cholangiopathy was 15 % versus 22 % (p = 0.73) and 5 % versus 2 % (p = 1), respectively. One-year patient survival and graft survival were 100 % versus 95 % (p = 0.18) and 90 % versus 95 % (p = 0.82), respectively. The authors concluded that the association of A-NRP + D-HOPE in DCD LT with prolonged WIT allowed achieving comparable outcomes to DBD LT. Moreover, these researchers stated that larger studies are needed to confirm these findings, refine the evaluation process, and establish when and by which modality machine perfusion is indicated in this setting.

The authors stated that drawbacks of this study included its retrospective, single-center design and limited sample sizes. Given the exploratory nature of this analysis, formal sample size calculation was not made. Furthermore, as the majority of DCD LTs were performed between 2020 and 2021, follow-up was shorter in DCD group. Although 6-months minimal follow-up should have allowed capturing the majority of biliary complications, late-onset complications could have been missed. These investigators were aware that an updated definition of functional WIT has been recently introduced; however, all cases included in this study were antecedent to its introduction and a retrospective re-calculation of functional WIT was not possible. Finally, as all grafts included in this study were treated with D-HOPE, these researchers could not examine the additional value of D-HOPE after A-NRP. It could be argued that use of machine perfusion could be

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omitted in selected cases, whereas additional viability assessment by normothermic machine perfusion could be indicated in others. In the authors' experience, use of D-HOPE has been systematic for grafts meeting all viability criteria during A-NRP, which were those included in this series. So far, use of normothermic machine perfusion has been limited to cases characterized by doubtful evaluation during A-NRP, or in which logistics constraints imposed prolonging preservation time. These investigators stated that well designed and appropriately powered randomized studies are needed to define when and by which modality machine perfusion after A-NRP is indicated in DCD LT.

Verstraeten and Jochmans (2022) stated that predicting organ viability before transplantation remains one of the most challenging and ambitious objectives in transplant surgery. Wait-list mortality is high while transplantable organs are discarded. Currently, approximately 20 % of deceased donor kidneys and livers are discarded because of "poor organ quality". Decisions to discard are still mainly a subjective judgement since there are only limited reliable tools predictive of outcome available. Organ perfusion technology has been posed as a platform for pre-transplant organ viability assessment. Markers of graft injury and function as well as perfusion parameters have been examined as possible viability markers during ex-situ hypothermic and normothermic perfusion. These investigators provided an overview of the available evidence for the use of kidney and liver perfusion as a tool to predict post-transplant outcomes. Although available evidence showed post-transplant outcomes could be predicted by both injury markers and perfusion parameters during hypothermic kidney perfusion, the predictive accuracy is too low to warrant clinical decision-making based upon these parameters alone. In liver, further evidence on the usefulness of hypothermic perfusion as a predictive tool is needed. Normothermic perfusion, during which the organ remains fully metabolically active, appeared a more promising platform for true viability assessment. Although these researchers do not yet fully understand "on-pump" organ behavior at normothermia, initial data in kidney and liver are promising. The authors concluded that although good quality evidence showed that injury markers and perfusion parameters during hypothermic kidney perfusion predicted graft outcome, these markers lack the predictive accuracy needed in clinical practice. These researchers stated that little is known regarding the association of liver perfusate injury markers and perfusion parameters during hypothermic perfusion and this deserves further investigation.

Liver Transplantation as a Rescue Therapy for Severe Neurologic Forms of Wilson Disease

Poujois and colleagues (2020) examined the effect of LT in patients with Wilson disease (WD) with severe neurologic worsening resistant to active chelation. French patients with WD who underwent LT for pure neurologic indication were retrospectively studied. Before LT and at the last follow-up, neurologic impairment was examined with the Unified Wilson's Disease Rating Scale (UWDRS) score, disability with the modified Rankin Scale (mRS) score, and hepatic function with the MELD score, together with the presence of a Kayser-Fleischer ring (KFR), brain MRI scores, and copper balance. The survival rate and disability at the last follow-up were the co-primary outcomes; evolution of KFR and brain MRI were the secondary outcomes. Prognosis factors were further assessed. A total of 18 patients had LT. All were highly dependent before LT (median mRS score of 5). Neurologic symptoms were severe (median UWDRS score of 105), dominated by dystonia and parkinsonism. The cumulated survival rate was 88.8 % at 1 year and 72.2 % at 3 and 5 years. At the last follow-up, 14 patients were alive. Their mRS and UWDRS scores improved ($p < 0.0001$ and $p = 0.0003$); 8 patients had a major improvement (78 % decrease of the UWDRS score), 4 a moderate one (41 % decrease), and 2 a stable status. KFR and brain MRI scores improved ($p = 0.0007$). Severe sepsis ($p = 0.011$) and intensive care unit (ICU) admission ($p = 0.001$) before LT were significantly associated with death. The authors concluded that LT is a rescue therapeutic option that should be carefully discussed in selected patients with neurologic WD resistant to anti-copper therapies (chelators or zinc salts) as it might allow patients to gain physical independency with a reasonable risk.

These researchers stated that although this study was the largest series ever published with the use of objective scores, it has several drawbacks: small number of patients ($n = 18$), retrospective evaluation of the patients, and lack of a control group. The rarity of the disease explained the limits as in France only 906 prevalent cases were identified. Moreover, they noted that the enrolled subjects were highly selected, large prospective studies are needed to validate these preliminary findings.

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Liver Transplantation for the Treatment of Hilar Cholangiocarcinoma

Moris and associates (2019) noted that hilar cholangiocarcinoma (hCCA) is a rare and aggressive malignancy with R0 resection being currently the only option for long-term survival. With the improvement in the outcomes of LT, the indications for LT have expanded to include other malignant tumors, such as hCCA. These researchers critically examined the outcomes of LT compared to resection with curative intent in patients with hCCA. They systematically searched the literature for articles published up to May 2018. The following algorithm was applied ((hilar cholangiocarcinoma) or (perihilar cholangiocarcinoma) or klatskin$ or (bile duct neoplasm) or cholangiocarcinoma) and (transplant$ OR graft$)). Neoadjuvant treatment with chemotherapy and radiation therapy was far more common in the LT group, with very few patients having received pre-operative therapy in the resection group ($p = 0.0005$). Moreover, length of hospital stay (LOS) was shorter after LT than after resection ($p < 0.00001$). In contrast, no difference was found between the 2 methods concerning post-operative mortality ($p = 0.57$). There was a trend towards longer OS following LT in comparison with resection. This was not obvious in the 1st year post-operatively, however, the advantage of LT over resection became obvious at 3 years after the operation ($p = 0.02$). The authors concluded that in non-disseminated unresectable tumors, LT appeared to have a non-inferior survival. In the same patients, neoadjuvant chemoradiotherapy and/or strict selection criteria may contribute to superior survival outcomes compared to curative-intent resection. However, these researchers stated that due to the scarcity of level 1 evidence, it remained unclear whether LT should be increasingly considered for technically resectable early stage hCCA.

The authors stated that this study had several drawbacks. First, all studies included were non-randomized, retrospective analyses and thus subject to the attendant biases. In particular, there was heterogeneity in terms of neoadjuvant treatment and whether tumors were associated with underlying primary sclerosing cholangitis (PSC) or arose de-novo. More specifically, only 1 study had available data for patients with and without PSC. The other studies did not provide sufficient granularity on this aspect. It should be noted that including PSC patients in this analysis may skew results towards longer OS with LT, since they are generally diagnosed at an earlier stage resulting in improved outcomes. Furthermore, the patients in the LT arm had locally unresectable disease, while the patients in the resection arm had resectable disease. In addition, there was heterogeneity among studies concerning staging and evaluating the extent of the disease. Furthermore ,these investigators noted that all median follow-up times were below 5 years and not many patients were followed-up up to 5 years post-operatively or more. Thus, conclusions regarding 5-year survival rates should be considered with caution. Finally, the small number of studies, and therefore patients, included in the analysis highlighted the necessity for follow-up comparative studies.

Machairas and colleagues (2020) stated that patients with hCCA have advanced disease at presentation, thus, curative therapeutic options are limited. Liver transplantation, in the case of unresectable disease, is theoretically an attractive option, as it offers the maximum resection margin and at the same time removes the underlying parenchymal liver disease. In the past years a number of studies have examined the potential beneficial role of neoadjuvant therapy followed by LT for treating patients with unresectable hCCA. In a systematic review, these investigators examined long-term outcomes of patients with hCCA undergoing LT. A systematic search of 4 electronic databases (Medline, Scopus, Google Scholar and ClinicalTrails.gov databases) was conducted for articles published between January 2000 and May 2019. A total of 13 studies with 698 patients were included in this systematic review. A proportion of 74.4 % of patients received combination of chemotherapy and radiation as a part of neoadjuvant therapy; 1-, 3- and 5-year OS rates ranged greatly among the included studies from 58 % to 92 %, 31 % to 80 % and 20 % to 74 %, respectively. Recurrence rates ranged from 16 % to 61 %, while peri-operative mortality ranged from 0 % to 25.5 %. Liver transplantation could provide acceptable long-term outcomes in the setting of neoadjuvant chemoradiation and strict patient selection criteria. The authors concluded that taking into account organ shortage, combined with the lack of level I evidence, more prospective, randomized trials are needed to establish certain indications, rigorous criteria and standardized protocols for LT in hCCA and provide the maximal potential benefits for these patients.

Scaffold-Based Transplantation

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Furuta and colleagues (2020) noted that OLT is the only treatment for end-stage liver failure; however, graft shortage impedes its applicability. Hence, studies examining alternative therapies are plenty. However, no study has comprehensively analyzed these therapies from different perspectives. These researchers summarized the current status of alternative transplantation therapies for OLT and to support future research. They carried out a systematic literature search using PubMed, Cochrane Library and Embase for articles published between January 2010 and 2018, using the following MeSH terms: "liver transplantation and cell", or "liver transplantation and differentiation", or "liver transplantation and organoid", or "liver transplantation and xenotransplantation". Various types of studies describing therapies to replace OLT were retrieved for full-text evaluation. Among them, these investigators selected articles including in-vivo transplantation. A total of 89 studies were selected. There were 3 principle forms of treatment for liver failure: Xeno-organ transplantation, scaffold-based transplantation, and cell transplantation. Xeno-organ transplantation was covered in 14 articles, scaffold-based transplantation was discussed in 22 articles, and cell transplantation was discussed in 53 articles. Various types of alternative therapies were discussed: Organ liver, 25 articles; adult hepatocytes, 31 articles; fetal hepatocytes, 3 articles; mesenchymal stem cells (MSCs), 25 articles; embryonic stem cells, 1 article; and induced pluripotent stem cells, 3articles and other sources. Clinical applications were discussed in 12 studies: Cell transplantation using hepatocytes in 4 studies, 5 studies using umbilical cord-derived MSCs, 3 studies using bone marrow-derived MSCs, and 2 studies using hematopoietic stem cells. The authors concluded that clinical applications are present only for cell transplantation. Scaffold-based transplantation is a comprehensive treatment combining organ and cell transplantations; future research on the clinical application of scaffold-based transplantation is expected.

Anti-Thrombotic Therapy for Prevention of Graft Thrombosis in Liver Transplant Recipients

Surianarayanan and colleagues (2021) noted that graft thrombosis is a well-recognized complication of solid organ transplantation and is one of the leading causes of graft failure. Currently, there are no standardized protocols for thrombo-prophylaxis. Many transplant units use unfractionated heparin (UFH) and fractionated heparins (low molecular weight heparin; LMWH) as prophylaxis for thrombosis. Antiplatelet agents such as aspirin are routinely used as prophylaxis of other thrombotic conditions and may have a role in preventing graft thrombosis. However, any pharmacological thrombo-prophylaxis comes with the theoretical risk of increasing the risk of major blood loss following transplant. In a Cochrane review, these investigators examined the benefits and harms of instituting thrombo-prophylaxis to patients undergoing solid organ transplantation. They searched the Cochrane Kidney and Transplant Register of Studies up to November 10, 2020 via contact with the Information Specialist using search terms relevant to this review. Studies in the Register are identified through searches of CENTRAL, Medline, and Embase, conference proceedings, the International Clinical Trials Register (ICTRP) Search Portal and ClinicalTrials.gov. These researchers included all RCTs and quasi-RCTs designed to examine interventions to prevent thrombosis in solid organ transplant recipients. All donor types were included (donor after circulatory (DCD) and brainstem death (DBD) and live transplantation). There was no upper age limit for recipients in the search. The results of the literature search were screened; and data collected by 2 independent authors. Dichotomous outcome results were expressed as RR with 95 % CI. Random effects models were used for data analysis. Risk of bias was independently examined by 2 authors using the risk of bias assessment tool. Confidence in the evidence was assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach.

These investigators identified 9 studies (712 participants); 7 (544 participants) included kidney transplant recipients, and studies included liver transplant recipients. They did not identify any study enrolling heart, lung, pancreas, bowel, or any other solid organ transplant recipient. Selection bias was high or unclear in 8 of the 9 studies; 5 studies were at high risk of bias for performance and/or detection bias; while attrition and reporting biases were in general low or unclear; 3 studies (180 participants) primarily examined heparinization in kidney transplantation. Only 2 studies reported on graft vessel thrombosis in kidney transplantation (144 participants). These small studies were at high risk of bias in several domains and reported only 2 graft thromboses between them; thus, it remained unclear whether heparin decreased the risk of early graft thrombosis or non-graft thrombosis (very low certainty). UFH may make little or no difference versus placebo to the rate of major bleeding events in kidney transplantation (3 studies, 155 participants: RR

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2.92, 95 % CI: 0.89 to 9.56; $I^2 = 0$ %; low certainty evidence). Sensitivity analysis using a fixed-effect model suggested that UFH may increase the risk of hemorrhagic events compared to placebo (RR 3.33, 95 % CI: 1.04 to 10.67, p = 0.04). Compared to control, any heparin (including LMWH) may make little or no difference to the number of major bleeding events (3 studies, 180 participants: RR 2.70, 95 % CI: 0.89 to 8.19; $I^2 = 0$ %; low certainty evidence) and had an unclear effect on risk of re-admission to intensive care (3 studies, 180 participants: RR 0.68, 95 % CI: 0.12 to 3.90, $I^2 = 45$ %; very low certainty evidence). The effect of heparin on other outcomes (including death, patient and graft survival, transfusion requirements) remained unclear (very low certainty evidence); 3 studies (144 participants) examined antiplatelet interventions in kidney transplantation: aspirin versus dipyridamole (1 study), and Lipo-PGE1 plus low-dose heparin to "control" in patients who had a diagnosis of acute rejection (2 studies). None of these reported on early graft thromboses. The effect of aspirin, dipyridamole and Lipo PGE1 plus low-dose heparin on any outcomes was unclear (very low certainty evidence); 2 studies (168 participants) examined interventions in liver transplants. One compared warfarin versus aspirin in patients with pre-existing portal vein thrombosis and the other examined plasmapheresis plus anti-coagulation. Both studies were abstract-only publications, had high risk of bias in several domains, and no outcomes could be meta-analyzed. Overall, the effect of any of these interventions on any of the outcomes remained unclear with no evidence to guide anti-thrombotic therapy in standard liver transplant recipients (very low certainty evidence).

The authors concluded that overall, there was a paucity of research in the field of graft thrombosis prevention. Due to a lack of high-quality evidence, it remained unclear whether any therapy was able to reduce the rate of early graft thrombosis in any type of solid organ transplant. UFH may increase the risk of major bleeding in kidney transplant recipients; however, this was based on low certainty evidence. There was no evidence from RCTs to guide anti-thrombotic strategies in liver, heart, lung, or other solid organ transplants. These researchers stated that further studies are needed to compare anti-coagulants, anti-platelets to placebo in solid organ transplantation; and these studies should focus on outcomes such as early graft thrombosis, major hemorrhagic complications, return to theatre, and patient/graft survival.

Hypothermic / Sub-Normothermic Machine Perfusion for Organ Preservation During Liver Transplantation

Bellini and associates (2019) stated that in order to match the current organ demand with organ availability from the donor pool, there has been a shift towards acceptance of extended criteria donors (ECD), often associated with longer ischemic times. Novel dynamic preservation techniques as hypothermic or normothermic machine perfusion (MP) are increasingly adopted, especially for organs from ECDs. In a systematic review and meta-analysis, these researchers compared the viability and incidence of re-perfusion injury in kidneys and livers preserved with MP versus static cold storage (SCS). They carried out a literature search between February and March 2019; Medline, Embase and Transplant Library were searched via OvidSP. The Cochrane Library and The Cochrane Central Register of Controlled Trials (CENTRAL) were also searched; English language filter was applied. The search generated a total of 10,585 studies, resulting in 30 papers for meta-analysis of kidneys and livers. Hypothermic MP (HMP) statistically significantly lowered the incidence of primary non-function (PNF, p = 0.003) and delayed graft function (DGF, p < 0.00001) in kidneys compared to SCS, but not its duration. No difference was also noted for serum creatinine or estimated glomerular filtration rate (eGFR) post-transplantation, but overall kidneys preserved with HMP had a significantly longer 1-year graft survival (OR: 1.61 95 % CI: 1.02 to 2.53, p = 0.04). Differently from kidneys where the graft survival was affected, there was no significant difference in PNF for livers stored using SCS for those preserved by HMP and normothermic machine perfusion (NMP). Machine perfusion demonstrated superior outcomes in early allograft dysfunction and post transplantation AST levels compared to SCS; however, only HMP was able to significantly decrease serum bilirubin and biliary stricture incidence compared to SCS. The authors concluded that MP improved DGF and 1-year graft survival in kidney transplantation; it appeared to mitigate early allograft dysfunction in livers; however, more studies are needed to prove the potential superiority of these novel technologies in relation to PNF in livers.

These researchers noted that contrary to the findings in the kidney, no difference in PNF was observed in livers preserved via HMP or NMP compared to SCS; in liver preservation both HMP and NMP had shown superior outcomes in mitigating early allograft dysfunction and post-transplantation AST levels compared to

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SCS; however, only HMP was able to significantly lower serum bilirubin and the incidence of biliary strictures, compared to SCS. Furthermore, the value of AST as an endpoint is controversial because there can be a release of AST in the perfusate during MP; thus, a more reliable marker should be considered in future studies. These conflicting results might be related to the relatively small number of RCTs with, therefore, no sufficient evidence to conclude a clear superiority of one modality compared to the other.

Horvath and colleagues (2021) noted that allograft ischemia during LT negatively affects the function of mitochondria, resulting in impairment of oxidative phosphorylation and compromised post-transplant recovery of the affected organ. Several preservation methods have been developed to improve donor organ quality; however, their effects on mitochondrial functions have not yet been compared. These researchers examined available evidence on mitochondrial effects of graft preservation methods in pre-clinical models of LT. In addition, they carried out a network meta-analysis to examine if any of these treatments provide a superior benefit, suggesting that they might be used on humans. These investigators carried out a systematic search using electronic databases (Embase, Medline (via PubMed), the Cochrane Central Register of Controlled Trials (CENTRAL) and Web of Science) for controlled animal studies using preservation methods for LT. The ATP content of the graft was the primary outcome, as this is an indicator overall mitochondrial function. Secondary outcomes were the respiratory activity of mitochondrial complexes, cytochrome c and aspartate aminotransferase (ALT) release. Both a random-effects model and the SYRCLE risk of bias analysis for animal studies were used. After a comprehensive search of the databases, 25 studies were enrolled in the analysis. Treatments that had the most significant protective effect on ATP content included HMP and sub-normothermic machine perfusion (SNMP) (MD = -1.0, 95 % CI: -2.3 to 0.3; and MD = -1.1, 95 % CI: -3.2 to 1.02), while the effects of warm ischemia (WI) without cold storage (WI) and normothermic machine perfusion (NMP) were less pronounced (MD = -1.8, 95 % CI: -2.9 to -0.7; and MD = -2.1 MD; CI: -4.6 to 0.4). The subgroup of SCS with shorter preservation time (less than 12 hours) yielded better results than SCS of greater than or equal to 12 hours, NMP and WI, in terms of ATP preservation and the respiratory capacity of complexes. HMP and SNMP stood out in terms of mitochondrial protection when compared to other treatments for LT in animals. The shorter storage time at lower temperatures, together with the dynamic preservation, provided superior protection for the grafts in terms of mitochondrial function. The authors concluded that additional clinical studies on human patients including marginal donors and longer ischemia times are needed to confirm any superiority of preservation methods with respect to mitochondrial function.

van Rijn and co-workers (2021) stated that transplantation of livers obtained from donors after circulatory death is associated with an increased risk of non-anastomotic biliary strictures. Hypothermic oxygenated machine perfusion of livers may reduce the incidence of biliary complications, but data from prospective, controlled studies are limited. In a controlled, multi-center trial, these researchers randomly assigned patients who were undergoing transplantation of a liver obtained from a donor after circulatory death to receive that liver either after hypothermic oxygenated machine perfusion (machine-perfusion group) or after conventional SCS alone (control group). The primary endpoint was the incidence of non-anastomotic biliary strictures within 6 months following transplantation; secondary endpoints included other graft-related and general complications. A total of 160 patients were enrolled, of whom 78 received a machine-perfused liver and 78 received a liver after SCS only (4 patients did not receive a liver in this trial). Non-anastomotic biliary strictures occurred in 6 % of the patients in the machine-perfusion group and in 18 % of those in the control group (RR, 0.36; 95 % CI: 0.14 to 0.94; p = 0.03). Post-reperfusion syndrome occurred in 12 % of the recipients of a machine-perfused liver and in 27 % of those in the control group (RR, 0.43; 95 % CI: 0.20 to 0.91). Early allograft dysfunction occurred in 26 % of the machine-perfused livers, as compared with 40 % of control livers (RR, 0.61; 95 % CI: 0.39 to 0.96). The cumulative number of treatments for non-anastomotic biliary strictures was lower by a factor of almost 4 following machine perfusion, as compared with control. The incidence of adverse events (AEs) was similar in the 2 groups. The authors concluded that hypothermic oxygenated machine perfusion resulted in a lower risk of non-anastomotic biliary strictures following the transplantation of livers obtained from donors after circulatory death than conventional SCS. Moreover, these researchers stated that in the present trial, machine perfusion did not have an effect on patient or graft survival; and given the high percentage of patients who survive following and the relatively low risk of graft loss, much larger trials are needed to detect an effect on these outcome measures.

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The authors stated that despite the restoration of ATP, hepatic metabolism remains suppressed and livers did not produce bile during this type of machine perfusion. Although the release of mitochondrial flavin mononucleotide into the perfusate has been correlated with hepatic function following LT, it remains unclear if this also predicts the risk of cholangiopathy. In contrast to normothermic machine perfusion, hypothermic machine perfusion is, therefore, currently not considered to be a tool for viability testing before transplantation. Furthermore, these investigators stated that whether hypothermic machine perfusion is beneficial in the transplantation of livers obtained from brain-dead donors is the subject of ongoing clinical trials.

Liver Transplantation for the Treatment of Methylmalonic Acidemia / Propionic Acidemia

Zhou and associates (2021) noted that the worldwide experience of LT in the treatment of propionic acidemia (PA) remains limited and fragmented. In a systematic review and meta-analysis, these investigators provided a comprehensive and quantitative understanding of post-transplant clinical outcomes in PA patients. Medline, Embase and the Cochrane Library databases were searched for studies focusing on PA patients who underwent LT. The pooled estimate rates and 95 % CIs were calculated using a random-effects model with Freeman-Tukey double arcsine transformation. A total of 21 studies involving 70 individuals were included. The pooled estimate rates were 0.95 (95 % CI: 0.80 to 1.00) for patient survival and 0.91 (95 % CI: 0.72 to 1.00) for allograft survival. The pooled estimate rates were 0.20 (95 % CI: 0.05 to 0.39) for rejection, 0.08 (95 % CI: 0.00 to 0.21) for hepatic artery thrombosis, 0.14 (95 % CI: 0.00 to 0.37) for cytomegalovirus / Epstein-Barr virus infection and 0.03 (95 % CI: 0.00 to 0.15) for biliary complications. The pooled estimate rates were 0.98 (95 % CI: 0.88 to 1.00) for metabolic stability, 1.00 (95 % CI: 0.79 to 1.00) for reversal of pre-existing cardiomyopathy and 0.97 (95 % CI: 0.78 to 1.00) for improvement of neurodevelopmental delay. A large proportion of patients achieved liberalization of protein intake post-transplant (pooled estimate rate 0.66 (95 % CI: 0.35 to 0.93). The authors concluded that despite the risk of transplant-related complications, LT was a viable therapeutic option in PA patients, with satisfactory survival rates and clinical outcomes. Moreover, these researchers stated that given the diversity in neurological assessment methods and the inconsistency in achievement of dietary protein liberalization across different studies, consensus on neurological evaluation methods and post-transplant protein intake is necessary; and further studies with longer-term clinical outcomes of LT for PA are needed.

Jiang and colleagues (2021) stated that LT and combined liver and kidney transplantation (CLKT) have been proposed as enzyme replacement therapies for the treatment of patients with methylmalonic aciduria (MMA). In a systematic review and meta-analysis, these investigators examined the available evidence on their safety and efficacy. Medline, Embase and Cochrane library were searched to identify studies that reported post-LT/CLKT clinical outcomes of MMA from their inception to February 1, 2020. The pooled rate was calculated using random-effects model with Freeman-Tukey double arcsine transformation method. A total of 32 studies involving 109 patients were included. The pooled estimate rates were 99.9 % (95 % CI: 95.3 to 100.0) for patient survival, 98.5 % (95 % CI: 91.5 to 100.0) for graft survival after LT/CLKT. The combined incidence of biliary, vascular complications and rejection were 0.2 % (95 % CI: 0.0 to 6.6), 7.7 % (95 % CI: 0.1 to 22.1) and 18.4 % (95 % CI: 4.6 to 36.3), respectively. The pooled estimate rates were 100.0 % (95 % CI: 99.4 to 100.0) for metabolic eradication, 61.5 % (95 % CI: 33.4 to 87.0) for normalization of kidney function. Chronic kidney disease (CKD) remission was more promising after CLKT (70.3 % versus 37.6 % in LT group). The pooled estimate rates for neurodevelopmental status improvement and protein intake liberalization were 52.0 % (95 % CI: 2.8 to 98.8) and 36.3 % (95 % CI: 6.3 to 71.7), respectively. The authors concluded that this 1st quantitative systematic review confirmed favorable survival outcomes and partially improved disease-related complications in transplanted MMA patients, although some results should be interpreted with caution. Moreover, these researchers stated that future studies with detailed description of long-term outcomes and consensus on neurodevelopmental evaluation method can help provide a more accurate picture.

The authors stated that this study had several drawbacks. First, owing to substantial heterogeneity in 2 endpoints (improvement in neurodevelopmental and protein intake), the statistical pooling may be misleading. Although subgroup analyses were carried out, these investigators were unable to explain the

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heterogeneity due to the paucity and invalidation of data. Furthermore, they could not conduct further analysis because the data provided by individual studies were limited. For other endpoints, the results were robust. Second, this systematic review was based mostly on case studies. Considering that case reports confirming clinical study conclusions have a higher probability of being published in a journal, a publication bias cannot be avoided. Third, the rather short period of the follow-ups of these studies, which made them unsuitable for evaluating the long-term outcome of transplantation. Based on the above evidences, these researchers suggested that future independent studies examining the efficacy of LT/CLKT for MMA should be based on a relatively large sample size with more detailed information, and examine the clinical outcomes using a standard tool.

Acute In-Patient Rehabilitation After Liver Transplantation

Mina et al (2022) noted that the indication and surgical complexity of OLT underscore the need for strategies to optimize the recovery for transplant recipients. In a systematic review, these researchers identified, evaluated, and synthesized the evidence examining the effect of in-patient rehabilitation for LT recipients and provided related practice recommendations. Health research databases were systematically reviewed for studies that included adults who received LT and participated in acute, post-transplant rehabilitation. Post-operative morbidity, mortality, hospital LOS, ICU LOS, and other markers of surgical recovery were extracted. Practice recommendations were provided by an international panel using GRADE. A total of 12 studies were included in the review (including 3,901 participants). Rehabilitation interventions varied widely in design and composition; however, details regarding intervention delivery were poorly described in general. The quality of evidence was rated as very low largely owing to “very serious” imprecision, poor reporting, and limited data from comparative studies. Overall, the studies suggested that in-patient rehabilitation for recipients of LT was safe, tolerable, and feasible, and may benefit functional outcomes. The authors concluded that 2 practice recommendations related to in-patient rehabilitation following LT were yielded from this review. First, it was safe, tolerable, and feasible. Second, it improved post-operative functional outcomes. Each of the recommendations was weak and supported by low-quality of evidence. These investigators stated that no recommendation could be made related to benefits or harms for clinical, physiological, and other outcomes. They stated that adequately powered and high-quality RCTs are needed in this area.

OrganOx Metra System for Transportation and Preservation of the Liver Prior to Transplantation

The OrganOx Metra System is a transportable medical device intended for normothermic perfusion of donor transplant livers for up to 24 hours. It is intended to be used to sustain donor livers destined for transplantation in a functioning state for a total preservation time of up to 12 hours. It is suitable for liver grafts from donors after brain death (DBD), or liver grafts from donors after circulatory death (DCD) of less than or equal to 40 years of age, with less than or equal to 20 mins of functional warm ischemic time (time from donor systolic blood pressure less than 50 mmHg), and macro-steatosis of less than or equal to 15 %, in a near-physiologic, normothermic and functioning state intended for a potential transplant recipient.

Gaurav et al (2022) noted that livers donated after circulatory death were associated with increased risk of primary non-function, poor function, and non-anastomotic strictures (NAS), leading to under-utilization. In a retrospective, single-center study, these researchers compared the outcomes of livers donated after circulatory death (DCD) and undergoing either in-situ normothermic regional perfusion (NRP) or ex-situ normothermic machine perfusion (NMP) with livers undergoing static cold storage (SCS). They prospectively collected data on 233 DCD liver transplants performed using SCS, NRP, or NMP between January 2013 and October 2020. A total of 97 SCS, 69 NRP, and 67 NMP DCD liver transplants were performed, with 6-month and 3-year transplant survival (graft survival non-censored for death) rates of 87 %, 94 %, 90 %, and 76 %, 90 %, and 76 %, respectively. NRP livers had a lower 6-month risk-adjusted Cox proportional hazard for transplant failure compared to SCS (hazard ratio [HR] 0.30, 95 % confidence interval [CI]: 0.08 to 1.05, $p = 0.06$). NRP and NMP livers had a risk-adjusted estimated reduction in the mean model for early allograft function score of 1.52 ($p < 0.0001$) and 1.19 ($p < 0.001$), respectively compared to SCS. Acute kidney injury (AKI) was more common with SCS (55 % versus 39 % NRP versus 40 % NMP; $p = 0.08$), with a lower risk-

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adjusted peak-to-baseline creatinine ratio in the NRP (p = 0.02). No NRP liver had clinically significant NAS in contrast to SCS (14 %) and NMP (11 %, p = 0.009), with lower risk-adjusted odds of overall NAS development compared to SCS (odds ratio [OR] = 0.2, 95 % CI: 0.06 to 0.72, p = 0.01). The authors concluded that NRP and NMP were associated with better early liver function compared to SCS, whereas NRP was associated with superior preservation of the biliary system. Moreover, these researchers stated that while the findings of this study were noteworthy, the recent data on hypothermic oxygenated machine perfusion (HOPE) highlighted the need to compare all new technologies in a randomized manner, and in particular to compare HOPE with NRP and NMP.

The authors stated that this study had several limitations. First, the retrospective design of the study, although the data analyzed were prospectively recorded. Second, NRP was predominantly undertaken in loco-regional donors (86 %), whereas NMP livers were often from remote hospitals outside the authors' region (78 %), after decline by at least 1 other center. These researchers acknowledged that this selection bias was a limitation, as it favored NRP for locally procured livers and did not favor NMP, which has less than ideal DCDs. Third, the long cold ischemia time (CIT) for all groups in the study, with that for the SCS group being 30 mins longer than NMP and NRP. To counter this, NRP had significantly longer asystolic and functional warm ischemia than the other 2 groups, with more of the NRP livers falling in the "high risk" category (SCS 43 %, NRP 64 %, NMP 37 %) as defined in a recent benchmarking paper. It is noteworthy that the superior outcomes of NRP livers were maintained after adjusting for CIT and other risk factors.

Furthermore, an UpToDate review on "Liver transplantation in adults: Deceased donor evaluation and selection" (Cotler, 2022) states that "Both hypothermic machine perfusion and normothermic ex-vivo liver evaluation are being studied as techniques to expand the donor pool by limiting the deleterious effects of cold ischemia on extended criteria grafts such as DCD liver grafts and livers with steatosis".

Epstein-Barr Viral Load Monitoring

Ruijter et al (2023) stated that primary infection with or reactivation of Epstein-Barr virus (EBV) can occur following LT and can lead to post-transplant lymphoproliferative disease (PTLD). In pediatric LT, an EBV-DNA viral load (EBV VL) monitoring strategy, including the reduction of immunosuppression, has led to a lower incidence of PTLD. For adult LT recipients with less primary infection and more EBV reactivation, it is unknown whether this strategy is effective. In a retrospective, cohort study, these researchers examined the effect of an EBV VL monitoring strategy on the incidence of PTLD following LT in adults. Adult recipients of 1st LT in Leiden between September 2003 and January 2017 with an EBV VL monitoring strategy formed the monitoring group (M1), recipients of 1st LT in Rotterdam between January 2003 and January 2017 without such a strategy formed the contemporary control group (C1), and those who had transplants in Leiden between September 1992 and September 2003 or Rotterdam between 1986 and January 2003 formed the historical control groups (M0 and C0, respectively). Measurements entailed influence of EBV VL monitoring on incidence of PTLD. After inverse probability of treatment weighting of the 4 groups to achieve a balance among the groups for important patient characteristics, differences within hospitals between the historical and recent era in cumulative incidences -- expressed as the number of events per 1,000 patients measured at 5-, 10-, and 15-year follow-up-showed fewer events in the contemporary era in both centers. This difference was considerably larger in the monitoring center, whereas the 95 % CI included the null value of 0 for point estimates. The authors concluded that monitoring EBV VL may reduce PTLD incidence after LT in adults; however, larger studies are needed. The drawbacks of this study included retrospective design, low statistical power, and incompletely balanced groups, and non-EBV PTLD could not be prevented.

Hepatic Glycogen Storage Disease

Beyzaei et al (2023) noted that LT is the choice of therapeutic option for end-stage hepatic glycogen storage disease (GSD) patients; however, reports regarding the long-term outcome of LT in these patients have remained controversial. In a systematic review and meta-analysis, observational studies published until December 31, 2021 were examined regarding the long-term outcome of LT in hepatic GSD patients. They carried out a literature search in the Medline/PubMed, Embase, Cochrane Library, Scopus and Web of

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Science Core Collection databases was performed. A total of 14 studies with 210 patients were included in this analysis. As the results showed, the pooled proportion of GSD patients with complications following LT (e.g., hemorrhagic shock, biliary complications, tacrolimus encephalopathy, chronic hepatitis, hepatic artery thrombosis, hepatic adenoma, sepsis, liver dysfunction, chronic rejection, ACR, and cytomegalovirus (CMV) infection) was 27.7 % (95 % CI: 20.42 to 35.67) without heterogeneity (I2 = 24.04 %), as calculated by the random-effect model. The pooled proportion of GSD patients with complications related to GSD following LT, including HCC, renal complication, muscle problems, delayed menarche, persistent neutropenia, pneumonitis, renal failure, and hepatic adenoma was 22.2 % (95 % CI: 7.97 to 40.01) with high heterogeneity (I2 = 82.47 %). Subgroup analysis including the age of patients (adult/pediatric), duration of follow-up, and type of donor was performed to examine the resources of heterogeneity. The authors concluded that according to their investigation and review analysis, most GSD patients showed significant outcome improvement following LT. Overall, these findings showed an excellent outcome of LT in GSD patients; however, further investigations are needed to confirm these findings.

Routine Use of Endobiliary Stent in Liver Transplantation

Elkomos et al (2023) stated that biliary complications are a significant cause of morbidity post-transplantation, and the routine use of biliary stents in LT to reduce these complications remains controversial. In a systematic review and meta-analysis, these investigators compared the incidence of biliary complications with and without the use of trans anastomotic biliary stent in LT. PubMed, Scopes, Web of Science, and Cochrane library were searched for eligible studies from inception to February 2022, and a systematic review and meta-analysis were performed to compare the incidence of biliary complications in the 2 groups. A total of 17 studies with 2,623 patients were included. The pooled results from the included studies demonstrated an equal rate of biliary complications (i.e., strictures, leaks and cholangitis) in stented and non-stented patients following LT. However, the cost and biliary intervention rates were higher in stented patients. Furthermore, the sub-group analysis showed no significant decrease in the incidence of biliary complications following the use of trans anastomotic biliary stent in LDLT, deceased donor LT (DDLT), Roux-en-Y hepaticojejunostomy (RYHJ), and duct-to-duct anastomosis, pediatric, and adult LT. The authors concluded that the routine use of endobiliary stent in LT provided no additional benefits; moreover, stented patients were at higher risk of needing multiple endoscopic retrograde cholangiopancreatographies (ERCPs). These researchers stated that further investigations are needed to compare the use of stent versus stentless in pediatric LT and RYHJ.

The authors stated that this study had several drawbacks. First, not all the included studies were RCTs. Second, the existence of significant heterogeneity in some outcomes could not be explained well enough by subgroup analysis. Third, none of the included studies compared the AR or the graft loss secondary to biliary complications between the 2 groups. Fourth, most of the studies were carried out in Asia and only few of them were performed in North America and Europe. Fifth, only 1 study was found comparing the use bio-absorbable stent versus stentless in biliary reconstruction.

Liver Elastography and Doppler Examination for Surveillance Following Liver Transplantation

An UpToDate review on “Liver transplantation in adults: Long-term management of transplant recipients” (Gaglio and Cotler, 2024) does not mention elastography / Fibrosure / FirboScan as a management option. However, it notes that “Complications of Immunosuppression -- An abdominal ultrasound with Doppler examination is performed in patients with cholestatic laboratory abnormalities or known biliary disease to evaluate for evidence of biliary obstruction and for hepatic artery thrombosis. We proceed to an MRCP or endoscopic retrograde cholangiopancreatography when there is clinical suspicion of a biliary stricture”.

High Mobility Group Box Protein 1 (HMGB1) Gene Polymorphisms Testing for Prediction of Morbidity and Mortality after Liver Transplantation

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Tsukiyama et al (2025) examined the effect of single-nucleotide polymorphisms (SNPs) in the high mobility group box 1 (HMGB1) gene on morbidity and mortality following LT. Among 120 LT recipients and their living donors, the genotypes of HMGB1, and the SNPs rs2249825, rs1045411, rs1412125, and rs1360485 were determined. There were no significant associations between these 4 SNPs and the incidence of rejection or mortality. However, the incidence of early allograft dysfunction (EAD) (n = 43), which presents as functional insufficiency within 1 week of LT, was significantly higher in recipients with the GC + CC allele of rs2249825 (n = 17/34) than in those with the GG allele (n = 26/86) (p = 0.044). Although the impact of donor HMGB1 SNPs on the incidence of EAD was not statistically significant, recipients with the GC + CC allele of rs2249825 who received liver grafts from donors with the same genotype had the highest incidence of EAD (p = 0.052). In contrast, the donor TC + CC allele of rs1412125 was an independent risk factor for the development of sepsis (n = 33) in LT recipient (OR = 3.05, 95 % CI: 1.18 to 7.87, p = 0.021). The authors concluded that the SNPs of the HMGB1 gene in either recipients or donors were not associated with mortality but influenced the incidence of EAD and sepsis, likely being a predictive biomarker for the risk of serious complications following LT.

Furthermore, UpToDate reviews on “Liver transplantation in adults: Initial and maintenance immunosuppression” (Vierling and Brandman, 2025), and “Liver transplantation in adults: Long-term management of transplant recipients” (Gaglio and Cotler, 2025) do not mention testing of high mobility group box 1 (HMGB1) gene as a management tool.

Glossary of Terms

Table: Glossary of Terms

Term	Definition
Orthotopic	Normal anatomical position

Appendix

A tool to calculate MELD score is available at the following website: MELD Calculator - OPTN.

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